

Harmonic Parity and Resolvent Identities

Exact Lerch generators, Stieltjes primitives,
and the quartic and higher Tornheim layers

Research continuation for the ProveIt project

11 October 2026

Abstract

We develop four exact continuations of the ProveIt program on polylogarithms, harmonic numbers, Gamma functions, and zeta derivatives. First, for an arbitrary polynomial numerator in generalized harmonic numbers, we classify the cancellation of every nonpositive even spectral principal part at the centered shift $1/2$. The criterion is invariance under the simultaneous reflection of all even harmonic tails; its necessity follows from a theorem on formal difference equations. We then evaluate every centered even moment of the square of trigamma as a rational number plus a rational multiple of $\zeta(3)$, and sum these values in a convergent polylogarithmic generator.

Second, a cotangent resolvent gives a completed formula for all generalized Stieltjes indices and all argument derivatives. We identify the entire local correction polynomial before coefficient extraction. Partial fractions yield every rational cotangent kernel bounded at infinity with no real pole. The same transform handles continuous polygamma orders and all balanced Gamma primitives, with an explicit conversion between analytic continuation in the order and pointwise endpoint finite parts.

Third, fixed nonpositive indices can be eliminated at arbitrary positions and depths in the entire relative harmonic interpolant. We prove a canonical finite polynomial normal form and a genuinely convergent Lerch generator for arbitrarily many zero indices surrounding one free order. Classical polynomial Stieltjes primitives are recorded with arbitrary endpoints and fully explicit coefficients.

Finally, every admissible quartic Tornheim ray is reduced to three specified coordinates, and cyclic quartic combinations retain only the diagonal coordinate. At all degrees we determine the exact homogeneous freedom under the stated symmetry, axis, and Euler-plane relations. An explicit deformation proves why diagonal data cease to determine the cyclic fifth layer. The article includes complete proofs, a source audit and integration notes, reproducible exact and numerical checks, and a concrete further-research agenda.

Source baseline. The manuscript and incoming reports are inspected at the fixed repository revision `c2cb285b64273e4cb2c8d7a03c68a4a99ac812d4`. Proof status and contributions are stated relative to that baseline. This article does not claim a proof of the unresolved S_6 or revised S_8 reductions.

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1 Scope, main results, and mathematical conventions

The objective is to obtain exact identities that survive the passage between spectral parameters, endpoint singularities, and ordinary special functions. Four aspects of the existing research collection are particularly productive: centered harmonic expansions, cotangent–Hurwitz transforms, entire relative harmonic interpolants, and holomorphic remainders of multivariate zeta functions. Each already has a substantial foundation in the manuscript or the incoming reports. The present article extends that foundation at specific open points.

1.1 The inspected baseline and the status of the questions

The source baseline is the repository revision `c2cb285b64273e4cb2c8d7a03c68a4a99ac812d4`, dated 11 October 2026 at 00:52:39 UTC. We inspected the relevant canonical manuscript chapters and the nine incoming archives listed in the accompanying source manifest. The most direct starting points are *Exact Resonant Identities* [2], *Independent Orders and Exact Reductions* [3], and *Cubic Tornheim Derivatives and Harmonic Stieltjes Constants* [4]. The original manuscript remains the reference for its established normalization and conjecture labels [1].

This is a targeted continuation and audit of those areas, not a claim to have checked every line of all manuscript volumes. In particular, the short S_6 reduction and the revised S_8 reduction remain conjectures. Previously rejected candidate vectors and previously reported corrections are distinguished from current candidates in Section 8.

The following table states the exact scope of the main advances. Resolution refers to the indicated mathematical question, with the domains and qualifications in the cited theorem.

Source question or direction	Result proved here	Location
Exact Resonant Q11: generalized harmonic cancellation	Necessary and sufficient centered involution criterion for all nonpositive even spectral points.	Theorem 2.1
Explicit centered values beyond raw H_n powers	Every trigamma-square even moment, its ordinary convergent subtracted sum, and a convergent polylogarithmic generator.	Theorems 3.1, 3.2
Exact Resonant Q8: a nonpolynomial cotangent family	The complete rational sector bounded at infinity, with the all-index endpoint correction.	Theorem 4.1; Corollary 4.2
Exact Resonant Q9: normalized primitive families	The resolvent transform of every balanced Gamma primitive and the continuous-order conversion formula.	Theorem 5.1; Corollary 5.2
Independent Orders: nonpositive-index reductions	Arbitrary positions and depths, a canonical finite normal form, and an analytic zero-decoration generator.	Theorems 6.1–6.3
Independent Orders and Cubic Harmonic: quartic rays	All quartic directions in three coordinates; cyclic directions in one diagonal coordinate.	Theorem 7.2; equation (7.8)
Higher directional relations	Exact formal kernel and cyclic-image dimensions for the named relations, and a fifth-order obstruction.	Theorem 7.4; Corollary 7.5

Table 1: Results relative to the pinned source questions. The primitive and higher-order statements resolve the specified families, rather than every possible extension proposed in the source.

1.2 A selection of exact identities

Three compact examples illustrate the different kinds of conclusion. The centered square identity

$$\sum_{n=0}^{\infty} \left(\left(n + \frac{1}{2} \right)^2 \psi'(n+1)^2 - 1 \right) = -\frac{1}{4} \zeta(3) - \frac{1}{6} \quad (1.1)$$

is an ordinary absolutely convergent series. It is the first nontrivial member of a complete all-even family, rather than a numerically recognized isolated value.

The pointwise endpoint finite part

$$\text{FP}_x \int_0^1 \frac{\psi'(x)}{\pi \cot(\pi x) - i\pi} dx = 1 - \gamma - \log(2\pi) + \frac{i\pi}{2} \quad (1.2)$$

has a boundary contribution equal to 1. Analytically continuing an integrated spectral family and then substituting the integer order without that contribution gives a different value. The all-index theorem locates this difference before any expansion.

For the Tornheim restriction $W_{a,b,c}(t) = T(at, bt, ct)$ and $\Lambda = W_{1,1,1}^{(4)}(0)$, a cyclic combination gives

$$\begin{aligned} W_{1,2,3}^{(4)}(0) + W_{2,3,1}^{(4)}(0) + W_{3,1,2}^{(4)}(0) - 36\Lambda \\ = -184 \log(2\pi) \zeta'''(0) + 156 \zeta''(0)^2 - 386 \zeta'''(0). \end{aligned} \quad (1.3)$$

The cancellation is an exact polynomial identity between directional coefficients. It does not require an assumption about relations among the individual constants.

1.3 Conventions and the meaning of a reduction

We use

$$\zeta(1+t, x) = \frac{1}{t} + \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(x) t^m, \quad \gamma_m = \gamma_m(1), \quad \gamma = \gamma_0,$$

and $\psi = \Gamma'/\Gamma$. Primes on a one-variable zeta function, or $\zeta^{(j)}(s, x)$ with the first argument displayed, mean derivatives in the spectral argument. Superscripts on ψ and $\gamma_m(x)$ mean derivatives in x . When confusion is possible, a square superscript $[j]$ denotes a polylogarithmic order derivative:

$$\text{Li}_s^{[j]}(z) = \partial_s^j \text{Li}_s(z).$$

All logarithms of positive real quantities are real. Complex powers use the specified branch; Hurwitz endpoints are initially in the right half-plane. Bernoulli numbers satisfy $B_1 = -1/2$. All branches required by the resolvent formulas are given explicitly in Section 4.

The term *reduction* is always relative to a stated target class. For the harmonic deletion theorem it means a finite linear combination with polynomial endpoint coefficients and only the surviving free orders. For a rational cotangent kernel it means finitely many polylogarithmic order derivatives at explicit arguments. For the quartic Tornheim layer it means three explicitly normalized coordinates, two of which are supplied with ordinary convergent representatives. It does not silently mean reduction to a conjectured algebra of familiar constants.

There are three distinct analytic operations in the proofs. A meromorphic continuation is uniquely determined from a convergence domain. A coordinate finite part extracts the constant term of a cutoff integral in its stated endpoint variable. A directional restriction of a meromorphic function is formed before taking its one-variable limit. These operations sometimes agree and sometimes differ by explicit terms. The article proves the needed compatibility at each use.

1.4 Proof inputs and attribution

The sufficient half-shift parity argument is elementary asymptotic algebra. Its converse uses the formal-Laurent version of the Adamczewski–Dreyfus–Hardouin theorem [7, Theorem 1.2]. This use is spelled out, including a proof that the relevant formal trigamma series is not rational. It supplies differential algebraic independence of functions and formal series; it says nothing about algebraic independence of Euler’s constant or zeta values.

The Mellin, Fourier, Bernoulli, Gamma, and polylogarithm expansions used below are classical [6]. The relative harmonic interpolant and the Tornheim polar normalization are inherited from the pinned incoming reports; the underlying one-endpoint interpolant is due to Ihara, Nakamura, and Yamamoto [8]. The continuous polygamma function and balanced negapolygamma functions are the established Espinosa–Moll functions [9]. Polynomial Hurwitz and Stieltjes primitives are also established in Espinosa–Moll [10] and the canonical manuscript. Our fully indexed arbitrary-endpoint version is included to make the new finite reduction calculus directly usable.

The centered trigamma-square family should also be distinguished from the weighted squares of *differences* of trigamma functions already evaluated in the incoming mixed-spectral report [5]. Both are exact identities; their summands, parameters, and generating mechanisms are different.

All results described as developments are developments relative to the inspected source baseline. The article makes no literature-wide priority claim. Proofs are analytic and algebraic; the accompanying programs provide finite exact checks and numerical diagnostics, not proof-assistant verification of the analytic theorems.

2 Exact parity of generalized harmonic numerators

Question Q11 in the pinned *Exact Resonant Identities* report [2] asks which polynomials in generalized harmonic numbers have no principal part at any nonpositive even spectral integer when the outer shift is $1/2$. The raw-power argument proves a sufficient condition for powers of H_n ; it does not itself classify arbitrary polynomials. This section gives a necessary and sufficient algebraic criterion. The necessity uses an established theorem on formal difference equations. It does not assume algebraic independence of Euler’s constant, zeta values, or numerical periods.

2.1 The centered involution and the classification theorem

Put $H_0^{(r)} = 0$, $H_n^{(r)} = \sum_{k=1}^n k^{-r}$, and $H_n = H_n^{(1)}$. For a fixed $R \geq 1$ and a polynomial $P \in \mathbb{C}[X_1, \dots, X_R]$, let

$$\mathcal{D}_P(s, a) = \sum_{n=0}^{\infty} \frac{P(H_n, H_n^{(2)}, \dots, H_n^{(R)})}{(n+a)^s}, \quad a > 0, \quad \Re s > 1. \quad (2.1)$$

These series converge normally in that half-plane; a polynomial numerator grows at most as a fixed power of $\log(n+2)$. All later values refer to the unique meromorphic continuation.

Define an involution on the polynomial ring by

$$\tau X_1 = X_1, \quad \tau X_r = \begin{cases} 2\zeta(r) - X_r, & r \geq 2 \text{ even}, \\ X_r, & r \geq 3 \text{ odd}. \end{cases} \quad (2.2)$$

The coefficients in $\mathbb{C}$ are fixed. Thus the transformation changes the sign of every even-order tail $\zeta(2j) - X_{2j}$ *simultaneously*.

Theorem 2.1 (Exact centered cancellation criterion). *For $P \in \mathbb{C}[X_1, \dots, X_R]$, the following conditions are equivalent:*

1. $\mathcal{D}_P(s, 1/2)$ is holomorphic at every $s = 0, -2, -4, \dots$
2. $\tau P = P$ as a polynomial with complex coefficients.
3. On writing $V_j = \zeta(2j) - X_{2j}$, every monomial of P has even total degree in the variables $V_1, V_2, \dots$

Equivalently, the complete algebra of such numerators is

$$\mathbb{C}[X_1, X_3, X_5, \dots, (\zeta(2i) - X_{2i})(\zeta(2j) - X_{2j}) : 1 \leq i \leq j \leq \lfloor R/2 \rfloor]. \quad (2.3)$$

Only variables whose index is at most R are retained.

The paired generators in (2.3) are a generating set, not an assertion that this algebra is freely generated. For example, $(V_i V_j)^2 = V_i^2 V_j^2$ is an algebraic relation among them.

Examples of admissible numerators are

$$H_n^p H_n^{(3)}, \quad (\zeta(2) - H_n^{(2)})^2, \quad H_n^p (\zeta(2) - H_n^{(2)})(\zeta(4) - H_n^{(4)}), \quad p \geq 0.$$

The raw polynomial $H_n^{(2)}$ does not qualify. In fact its principal part at $s = 0$ is $-1/s$, since $H_n^{(2)} = \zeta(2) - (n + 1/2)^{-1} + O(n^{-3})$. Multiplying two even-order tails changes this outcome by making their full centered expansion even.

2.2 A finite formula for every principal part

Let L be an independent formal symbol, z a formal variable, and

$$A(z, a) = - \sum_{k \geq 1} \frac{B_k(a)}{k} z^k, \quad T_1(z, L; a) = L + \gamma + A(z, a). \quad (2.4)$$

For $r \geq 2$ define

$$T_r(z; a) = \zeta(r) - \frac{z^{r-1}}{r-1} - \frac{1}{r-1} \sum_{k \geq 1} \binom{k+r-2}{r-2} B_k(a) z^{k+r-1}. \quad (2.5)$$

These are formal asymptotic series; convergence in z is not needed. Set

$$P(T_1, T_2, \dots, T_R) = \sum_{j \geq 0} q_{P,j}(L; a) z^j, \quad q_{P,j}(L; a) = \sum_{\ell \geq 0} q_{P,j,\ell}(a) L^\ell. \quad (2.6)$$

Every coefficient requires only finitely many Bernoulli polynomials and finite polynomial operations.

Proposition 2.2 (Generalized-harmonic principal coefficients). *The function (2.1) continues meromorphically to the entire s -plane. Its possible poles are $1, 0, -1, -2, \dots$. At $s = 1 - j$ its complete principal part is*

$$\sum_{\ell \geq 0} \frac{\ell! q_{P,j,\ell}(a)}{(s + j - 1)^{\ell+1}}. \quad (2.7)$$

The sum is finite. In particular, this point is regular if and only if $q_{P,j}(L; a)$ is the zero polynomial in L .

Proof. Writing $x = n + a$, the ordinary shifted digamma expansion and its fixed argument derivatives give the expansions T_r above, with $z = x^{-1}$ and $L = \log x$; see the classical Gamma and polygamma formulas in [6, §§5.11, 5.15]. For $r \geq 2$ one can also obtain (2.5) directly by differentiating

$$H_n = \psi(n+1) + \gamma, \quad \zeta(r) - H_n^{(r)} = \frac{(-1)^r}{(r-1)!} \psi^{(r-1)}(n+1).$$

At every fixed truncation order J , the polynomial substitution has remainder

$$O(x^{-J-1}(1 + \log^d x))$$

for some fixed d . Subtract those finitely many terms from the Dirichlet series. The remaining series converges normally in $\Re s > -J$ on compact subsets. The subtracted terms are

$$\sum_{j=0}^J \sum_{\ell} q_{P,j,\ell}(a) (-1)^\ell \partial_s^\ell \zeta(s+j, a).$$

Their only principal contribution at $s = 1 - j$ is (2.7). Taking arbitrarily large J proves the continuation. Distinct powers $(s+j-1)^{-\ell-1}$ cannot cancel one another, proving the last assertion. $\square$

At $a = 1/2$, all odd Bernoulli polynomials vanish. Consequently

$$T_1(-z, L; 1/2) = T_1(z, L; 1/2), \quad T_r(-z; 1/2) = \begin{cases} 2\zeta(r) - T_r(z; 1/2), & r \text{ even,} \\ T_r(z; 1/2), & r \text{ odd.} \end{cases} \quad (2.8)$$

This proves the sufficiency of $P = \tau P$. The necessity requires showing that no nonzero polynomial is annihilated by the formal substitution. A finite list of verified asymptotic coefficients could not establish that fact.

2.3 Why the formal substitution is injective

We use the following established result in exactly its formal Laurent setting. Adamczewski, Dreyfus, and Hardouin [7, Theorem 1.2, case $\mathbf{S}_\infty$] prove that a formal Laurent series $f \in \mathbb{C}((x^{-1}))$ satisfying a homogeneous linear difference equation over $\mathbb{C}(x)$ for the shift $x \mapsto x+1$ is either rational or differentially transcendental over $\mathbb{C}(x)$. The latter means that every finite tuple $f, f', \dots, f^{(q)}$ is algebraically independent over $\mathbb{C}(x)$.

Lemma 2.3 (Independence of the centered formal harmonic tails). *The substitution*

$$\mathbb{C}[X_1, \dots, X_R] \longrightarrow \mathbb{C}[L][[z]], \quad X_r \longmapsto T_r(z, L; 1/2),$$

is injective.

Proof. Let

$$f(x) = \frac{1}{x} + \sum_{k \geq 1} B_{2k}(1/2) x^{-2k-1} \in \mathbb{C}((x^{-1})). \quad (2.9)$$

This is the formal expansion of $\psi'(x+1/2)$. The trigamma shift identity, or the Bernoulli generating series, gives

$$f(x+1) - f(x) = -\frac{1}{(x+1/2)^2}. \quad (2.10)$$

In particular it satisfies the homogeneous second-order equation

$$f(x+2) - (1+c(x))f(x+1) + c(x)f(x) = 0, \quad c(x) = \frac{(x+1/2)^2}{(x+3/2)^2}. \quad (2.11)$$

There is no rational solution of (2.10). Indeed, in any nonempty finite set of poles of a rational function belonging to one coset modulo $\mathbb{Z}$, choose the first and last pole in the integer ordering. Its forward difference has an uncanceled pole immediately before the first and another at the last. These poles are distinct. Thus a rational forward difference cannot have precisely one pole in such a coset. The right side of (2.10) has exactly one pole. A rational function with no finite poles is a polynomial, whose forward difference is polynomial, and is also impossible.

The cited difference-equation theorem applied to (2.11) now shows that $f, f', \dots, f^{(R-2)}$ are algebraically independent over $\mathbb{C}(x)$ when $R \geq 2$. For $r \geq 2$, the substitution for X_r is the affine expression

$$T_r = \zeta(r) - \frac{(-1)^r}{(r-1)!} f^{(r-2)}.$$

These tails are therefore algebraically independent, even over $\mathbb{C}(x)$. Finally $T_1 = L + \gamma + A(z, 1/2)$ has coefficient one in the independent formal symbol L , and the other T_r contain no L . If a nonzero polynomial in X_1 has degree d and nonzero leading coefficient $a_d(X_2, \dots, X_R)$, the coefficient of L^d after substitution is the nonzero substituted value of a_d . This proves injectivity, including the case $R = 1$ directly. $\square$

Proof of Theorem 2.1. By Proposition 2.2, regularity at every $s = -2m$ is equivalent to the vanishing of every odd coefficient $q_{P, 2m+1}$. Equivalently, the complete formal substitution is even in z while L is held fixed. By (2.8), this says that the substitution annihilates $P - \tau P$. Lemma 2.3 therefore gives $P - \tau P = 0$.

Writing the even-index variables as $\zeta(2j) - V_j$ makes τ the simultaneous sign change of all the V_j . The invariant polynomials are exactly those with even total degree in these variables. Every such monomial can be grouped into pairs $V_i V_j$, proving the stated finite generating description. $\square$

Corollary 2.4 (Complementary pole lattices). *Write $P = P_+ + P_-$, where $P_{\pm} = (P \pm \tau P)/2$. Then $\mathcal{D}_{P_+}(s, 1/2)$ has possible poles only at $1, -1, -3, \dots$, while $\mathcal{D}_{P_-}(s, 1/2)$ has possible poles only at $0, -2, -4, \dots$. Moreover, if $P \neq 0$, $\mathcal{D}_P(s, 1/2)$ is not entire.*

Proof. The corresponding formal expansions have only even and only odd powers of z , respectively. This gives the two pole lists by (2.7). If the whole Dirichlet function were entire, all its coefficient polynomials $q_{P,j}$ would vanish. Injectivity would then force $P = 0$. $\square$

This is a statement about identities of functions and formal series, not about algebraic independence of their special numerical values. The criterion concerns cancellation at *every* point of the specified parity. At any finite list of spectral points, additional accidental cancellations can occur and are decided by the finite Bernoulli conditions (2.7).

3 All centered trigamma-square moments and a polylogarithmic generator

The invariant numerator $(\zeta(2) - H_n^{(2)})^2$ provides an explicit family beyond raw powers of H_n . Define

$$A_n = \zeta(2) - H_n^{(2)} = \psi'(n+1), \quad \mathcal{T}(s) = \sum_{n=0}^{\infty} \frac{A_n^2}{(n+1/2)^s}. \quad (3.1)$$

The defining series already converges for $\Re s > -1$. By the preceding theorem it is regular at every nonpositive even integer. All its values there reduce to a rational number plus a rational multiple of $\zeta(3)$.

Theorem 3.1 (Every nonpositive even trigamma-square value). *For every integer $r \geq 0$,*

$$\begin{aligned} \mathcal{T}(-2r) &= 3B_{2r}(1/2)\zeta(3) \\ &\quad - \frac{1}{2} \sum_{j=1}^r \binom{2r}{2j} \frac{2j-1}{2j+1} B_{2r-2j}(1/2) B_{2j-2}. \end{aligned} \quad (3.2)$$

The Bernoulli number convention is $B_1 = -1/2$, and the sum is empty when $r = 0$.

For example,

$$\begin{aligned} \mathcal{T}(0) &= 3\zeta(3), \\ \mathcal{T}(-2) &= -\frac{1}{4}\zeta(3) - \frac{1}{6}, \\ \mathcal{T}(-4) &= \frac{7}{80}\zeta(3) + \frac{1}{30}, \\ \mathcal{T}(-6) &= -\frac{31}{448}\zeta(3) + \frac{1}{672}, \\ \mathcal{T}(-8) &= \frac{127}{1280}\zeta(3) - \frac{17}{540}. \end{aligned} \quad (3.3)$$

The first value is the classical Euler sum. The theorem supplies a single finite formula for all subsequent centered moments.

3.1 Equivalent identities of absolutely convergent ordinary sums

Let

$$c_k = \sum_{i=0}^{k-1} B_{2i}(1/2) B_{2k-2-2i}(1/2), \quad k \geq 1. \quad (3.4)$$

Thus $c_1 = 1$, $c_2 = -1/6$, $c_3 = 47/720$, and $c_4 = -257/5040$. Squaring (2.9) gives $A_n^2 \sim \sum_{k \geq 1} c_k (n + 1/2)^{-2k}$. For every $r \geq 0$ the exact ordinary convergent identity is

$$\sum_{n=0}^{\infty} \left\{ (n + 1/2)^{2r} \psi'(n + 1)^2 - \sum_{k=1}^r c_k (n + 1/2)^{2r-2k} \right\} = \mathcal{T}(-2r), \quad (3.5)$$

where the right side is explicitly (3.2). The summands after subtraction are $O(n^{-2})$.

For instance,

$$\sum_{n=0}^{\infty} \left((n + 1/2)^2 \psi'(n + 1)^2 - 1 \right) = -\frac{1}{4}\zeta(3) - \frac{1}{6},$$

and

$$\sum_{n=0}^{\infty} \left((n + 1/2)^4 \psi'(n + 1)^2 - (n + 1/2)^2 + \frac{1}{6} \right) = \frac{7}{80}\zeta(3) + \frac{1}{30}.$$

These statements need no prescription for summing a divergent series. To prove (3.5), subtract the displayed asymptotic terms in the half-plane of convergence and continue. Their Dirichlet sums at $s = -2r$ are $c_k \zeta(2k - 2r, 1/2) = 0$, since $\zeta(-2m, 1/2) = 0$ for $m \geq 0$.

3.2 Proof of the Bernoulli formula by finite summation

For a sequence admitting an expansion in powers of N and $\log N$, write CT_N for its coefficient of $N^0(\log N)^0$. Only finite exact identities followed by known asymptotic expansions will be used here. Put

$$S_r(k) = \sum_{n=0}^{k-1} (n+1/2)^{2r} = \frac{B_{2r+1}(k+1/2)}{2r+1} = \sum_{j=0}^r b_{r,j} k^{2j+1}, \quad (3.6)$$

where

$$b_{r,j} = \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2). \quad (3.7)$$

The Bernoulli polynomial $B_{2r+1}(1/2)$ is zero, so this is an odd polynomial in k and has zero constant term.

Proof of Theorem 3.1. First, (3.5) and (3.6) show that

$$\mathcal{T}(-2r) = \text{CT}_N \sum_{n=0}^{N-1} (n+1/2)^{2r} A_n^2. \quad (3.8)$$

Indeed every subtracted even power has partial sum with zero constant term in N , and the convergent remainder tends to the stated value.

The identity $A_{k-1} = A_k + k^{-2}$ gives finite summation by parts:

$$\sum_{n=0}^{N-1} (n+1/2)^{2r} A_n^2 = \sum_{k=1}^N S_r(k) \left(\frac{2A_k}{k^2} + \frac{1}{k^4} \right) + S_r(N) A_N^2. \quad (3.9)$$

The coefficient $b_{r,0}$ of k contributes

$$\text{CT}_N \left(2 \sum_{k=1}^N \frac{A_k}{k} + H_N^{(3)} + N A_N^2 \right) = 3\zeta(3).$$

Here $\sum_{k \geq 1} A_k/k = \zeta(2, 1) = \zeta(3)$ is the classical Euler identity and $N A_N^2 \rightarrow 0$. For completeness, the Euler identity follows from

$$\zeta(2, 1) = \frac{1}{2} \int_0^1 \frac{\log^2(1-t)}{t} dt = \frac{1}{2} \int_0^\infty \frac{u^2}{e^u - 1} du = \zeta(3).$$

For $j \geq 1$, it remains to compute the constant of

$$E_j(N) = 2 \sum_{k=1}^N k^{2j-1} A_k + \sum_{k=1}^N k^{2j-3} + N^{2j+1} A_N^2. \quad (3.10)$$

Let

$$F_j(N) = \sum_{k=1}^N k^{2j-1} = \frac{B_{2j}(N+1) - B_{2j}}{2j}.$$

Changing the order in a finite sum gives

$$\sum_{k=1}^N k^{2j-1} A_k = A_N F_j(N) + \sum_{\ell=1}^N \frac{F_j(\ell-1)}{\ell^2}. \quad (3.11)$$

For $j \geq 2$, the second summand is a sum of polynomial powers of ℓ and has zero constant term as a polynomial in N ; $B_{2j-1} = 0$ removes a possible $1/\ell$ term. The other power sum in (3.10)

also has zero constant. For $j = 1$ the second sum in (3.11) equals $(N - H_N)/2$; its harmonic term cancels exactly against $\sum k^{-1}$ in (3.10). In all cases,

$$\text{CT}_N E_j(N) = 2 \text{CT}_N(A_N F_j(N)) + \text{CT}_N(N^{2j+1} A_N^2). \quad (3.12)$$

Now use the uncentered tail expansion

$$A_N = \frac{1}{N} - \frac{1}{2N^2} + \sum_{k \geq 1} \frac{B_{2k}}{N^{2k+1}}.$$

The finite coefficient calculation gives

$$\text{CT}_N(A_N F_j(N)) = \frac{3-2j}{4} B_{2j-2}, \quad \text{CT}_N(N^{2j+1} A_N^2) = -B_{2j-2}. \quad (3.13)$$

To check the first identity, the coefficient of N^ℓ in F_j is $\binom{2j}{\ell} B_{2j-\ell}(1)/(2j)$. Only $\ell = 2$ and $\ell = 2j - 1$ can pair with nonzero inverse coefficients of A_N when $j \geq 2$; their contributions are $-(2j-1)B_{2j-2}/4$ and $B_{2j-2}/2$. The case $j = 1$ is the direct product with $F_1(N) = (N^2 + N)/2$ and gives $1/4$. For the second identity, an odd inverse exponent in A_N^2 arises only by multiplying $-1/(2N^2)$ by the appropriate odd inverse power; this gives $-B_{2j-2}$, also at $j = 1$.

Combining (3.12) and (3.13) yields

$$\text{CT}_N E_j(N) = -\frac{2j-1}{2} B_{2j-2}.$$

Multiplication by $b_{r,j}$, followed by the $j = 0$ contribution, proves (3.2). $\square$

3.3 A convergent generator involving classical polylogarithms

The complete sequence has an analytic exponential generating function. Let

$$\mathcal{G}(t) = \sum_{r=0}^{\infty} \mathcal{T}(-2r) \frac{t^{2r}}{(2r)!}.$$

Theorem 3.2 (Polylogarithmic moment generator). *The series $\mathcal{G}$ converges for $|t| < 2\pi$ and equals*

$$\mathcal{G}(t) = \frac{t}{2 \sinh(t/2)} \left[3\zeta(3) - \frac{1}{4t} \int_0^t (t-u) u^2 \coth(u/2) du \right]. \quad (3.14)$$

The singularities at zero in this expression are removable. For real $0 < t < 2\pi$, an equivalent formula in classical polylogarithms is

$$\mathcal{G}(t) = \frac{t}{2 \sinh(t/2)} \left[2\zeta(3) - \frac{t^3}{48} - \frac{t}{2} \text{Li}_2(e^{-t}) - 2 \text{Li}_3(e^{-t}) + \frac{3(\zeta(4) - \text{Li}_4(e^{-t}))}{t} \right]. \quad (3.15)$$

Its continuation specified by (3.14) is holomorphic and even throughout $|t| < 2\pi$.

Proof. The Bernoulli generating functions give

$$\frac{t}{2 \sinh(t/2)} = \sum_{r \geq 0} B_{2r}(1/2) \frac{t^{2r}}{(2r)!}, \quad \frac{u}{2} \coth(u/2) = \sum_{k \geq 0} B_{2k} \frac{u^{2k}}{(2k)!}.$$

Both converge in the disk of radius 2π . Integrating the second series twice gives

$$\frac{1}{2t} \int_0^t (t-u) u^2 \coth(u/2) du = \sum_{j \geq 1} \frac{2j-1}{2j+1} B_{2j-2} \frac{t^{2j}}{(2j)!}.$$

Thus (3.14) is exactly the binomial convolution in (3.2). This proves convergence, analyticity and the integral formula. In particular all apparent odd or logarithmic terms in any branch representation must cancel.

For positive t , expand $(e^u - 1)^{-1}$ in its convergent positive exponential series. Direct integration, justified for the integrable factor $(t - u)u^2$, gives

$$\int_0^t \frac{(t - u)u^2}{e^u - 1} du = 2t\zeta(3) - 6\zeta(4) + t^2 \text{Li}_2(e^{-t}) + 4t \text{Li}_3(e^{-t}) + 6 \text{Li}_4(e^{-t}).$$

Finally $\coth(u/2) = 1 + 2/(e^u - 1)$ and $\int_0^t (t - u)u^2 du = t^4/12$ give (3.15). □

3.4 Verification, audit scope, and further exact questions

The accompanying `verify_harmonic_parity.py` implements the formal involution, verifies the centered trigamma difference equation, and checks the two finite Bernoulli constant calculations independently for eight values of j . It evaluates seven spectral values directly from finite trigamma sums with independently truncated asymptotic tails at two cutoffs, using 110 decimal digits. The largest discrepancy against (3.2) is less than 1.6×10^{-85} . At three positive parameters it compares the polylogarithmic generator both with the Bernoulli moment series and with ordinary quadrature of (3.14). These are floating-point diagnostics, not interval enclosures. The proofs do not depend on their precision.

The selected harmonic statements in the pinned report were consistent with these checks. No new false formula was identified in that selected source. Its Q11 warning was correct: raw-power parity alone does not apply to an arbitrary higher-harmonic numerator. The result here resolves that specific question by supplying the exact invariant ring; it does not label the earlier warning an error.

The following are concrete extensions beyond the proved results.

1. Evaluate every centered moment of $(\zeta(2i) - H_n^{(2i)})(\zeta(2j) - H_n^{(2j)})$ by a uniform finite Bernoulli formula, and determine the precise convergent multiple-zeta coordinates that remain. The current trigamma-square result is $i = j = 1$.
2. For invariant products of three or more tail pairs, seek a polylogarithmic generator analogous to (3.15) and determine when the resulting multiple zeta values reduce to ordinary zeta products.
3. Classify cancellation at a *finite* prescribed set of spectral integers by the finite coefficient map (2.7). Its kernel can be larger than the invariant ring and is a tractable exact linear problem in any bounded polynomial space.
4. Derive exact spectral-derivative identities for $\mathcal{T}$ at its regular even points. Such derivatives retain a global remainder and do not follow by formally differentiating the discrete index r in (3.2).
5. Determine an elementary proof of the injectivity lemma tailored to the additive trigamma difference equation. This would remove the use of the general difference Galois theorem, while leaving the current unconditional classification unchanged.

4 Cotangent resolvents: a nonpolynomial Stieltjes generator

Question Q8 of the incoming *Exact Resonant Identities* report [2] asks for a nonpolynomial extension of its cotangent moment calculus, including the local subtraction before parameter

expansion. We give such an extension for every rational function of the cotangent bounded at infinity and with no real pole. A single resolvent supplies the answer; differentiation gives all repeated poles, and partial fractions give the general rational case. The answer involves ordinary polylogarithms and their order derivatives. The Fourier identity used in the proof is classical. The contribution here is the completed rational-kernel identity, its precise endpoint correction at every Stieltjes and argument-derivative order, and its compatibility with continuous polygamma orders and primitives.

4.1 Coordinates, branches, and finite parts

Write

$$K(x) = \pi \cot(\pi x), \quad R_z(x) = \frac{1}{K(x) - z}, \quad 0 < x < 1, \quad z \in \mathbb{C} \setminus \mathbb{R}. \quad (4.1)$$

Although K has endpoint poles, R_z extends to a smooth periodic function, with $R_z(0) = R_z(1) = 0$. On either open half-plane put

$$\epsilon = \operatorname{sgn}(\Im z), \quad a = z + \epsilon\pi i, \quad q = \frac{z - \epsilon\pi i}{z + \epsilon\pi i}, \quad c = \frac{2\epsilon\pi i}{a^2}, \quad \sigma = -2\epsilon\pi i. \quad (4.2)$$

Thus $|q| < 1$, and throughout the section

$$\operatorname{Log} \sigma = \log(2\pi) - \epsilon i\pi/2.$$

The letter a in this section is only the denominator in (4.2); it is not a Hurwitz endpoint. To remove an otherwise unnecessary exceptional case, define

$$\Phi_*(s, q) = \frac{\operatorname{Li}_s(q)}{q} = \sum_{n \geq 1} \frac{q^{n-1}}{n^s}, \quad \Phi_*(s, 0) = 1. \quad (4.3)$$

This function is entire in s and holomorphic for $|q| < 1$. Its order derivatives at $q = 0$ are zero except at order zero.

We use the conventional generalized Stieltjes functions

$$\zeta(1-s, x) = -\frac{1}{s} + \sum_{m \geq 0} \gamma_m(x) \frac{s^m}{m!}, \quad \gamma_0(x) = -\psi(x). \quad (4.4)$$

A superscript (p) on γ_m or ψ means an argument derivative. A superscript $[j]$ on Li_s or Φ_* means an order derivative. These two operations must be distinguished.

Every finite part below is taken in the original variable x : $\operatorname{FP}_x \int_0^1 f(x) dx$ is the coefficient of $\varepsilon^0(\log \varepsilon)^0$ in $\int_\varepsilon^{1-\varepsilon} f(x) dx$. Equivalently, subtract the finitely many nonintegrable endpoint terms, integrate the remainder, and add the constant terms of the explicit subtractions. For the resolvent integrals only the endpoint $x = 0$ can be singular. The Stieltjes shift formula gives

$$\gamma_m(x) = \frac{\log^m x}{x} + \gamma_m(x+1). \quad (4.5)$$

Consequently all the finite parts used here exist. In particular,

$$M_{m,p}(z) := \operatorname{FP}_x \int_0^1 \gamma_m^{(p)}(x) R_z(x) dx$$

is an ordinary convergent integral when $p = 0$.

4.2 The complete all-index identity

Set

$$B_\epsilon(s) = \Gamma(1+s)\sigma^{-s}, \quad E_p(s) = \prod_{j=1}^p \left(1 - \frac{s}{j}\right), \quad E_0(s) = 1. \quad (4.6)$$

Theorem 4.1 (Completed cotangent–Stieltjes resolvent). *For all integers $m, p \geq 0$ and $z \notin \mathbb{R}$,*

$$M_{m,p}(z) = m!c\sigma^p[s^{m+1}] \{B_\epsilon(s)\Phi_*(s-p, q) - E_p(s)\Phi_*(-p, q)\}. \quad (4.7)$$

The expression in braces vanishes at $s = 0$. The identity is holomorphic in z on each half-plane, including $z = \epsilon i\pi$. For $p \geq 1$ the polynomial E_p is the full local endpoint correction. Its contribution vanishes whenever $m \geq p$.

The identity can be expanded without any coefficient notation. Let $g_j = B_\epsilon^{(j)}(0)$ and $e_{p,j} = E_p^{(j)}(0)$. Then

$$M_{m,p}(z) = \frac{c\sigma^p}{m+1} \left\{ \sum_{j=0}^{m+1} \binom{m+1}{j} g_{m+1-j} \Phi_*^{[j]}(-p, q) - e_{p,m+1} \Phi_*(-p, q) \right\}. \quad (4.8)$$

The coefficients g_j are complete exponential Bell polynomials with logarithmic derivatives

$$b_1 = -\gamma - \text{Log } \sigma, \quad b_j = (-1)^j (j-1)! \zeta(j) \quad (j \geq 2),$$

so $g_0 = 1$, $g_1 = b_1$, $g_2 = b_1^2 + \zeta(2)$, and $g_3 = b_1^3 + 3b_1\zeta(2) - 2\zeta(3)$. Likewise

$$e_{p,j} = (-1)^j j! e_j \left(1, \frac{1}{2}, \dots, \frac{1}{p}\right), \quad (4.9)$$

where e_j on the right is the elementary symmetric polynomial; it is zero for $j > p$. Thus the contact terms use ordinary harmonic numbers and their finite Bell-polynomial combinations.

Proof of Theorem 4.1. It is enough to prove the upper-half-plane case; reversing the Fourier signs gives the other case with exactly the branches in (4.2). Put $w = e^{2\pi i x}$. Direct algebra gives the uniformly convergent Fourier expansion

$$R_z(x) = -\frac{1}{a} + c \sum_{n \geq 1} q^{n-1} e^{2\pi i n x}. \quad (4.10)$$

For example, at $z = i\pi$ this reduces to $(e^{2\pi i x} - 1)/(2\pi i)$. The classical Hurwitz Fourier formula [6, §25.11] implies, initially in a sufficiently large right half-plane,

$$\int_0^1 \zeta(1-s, x) dx = 0, \quad \int_0^1 \zeta(1-s, x) e^{2\pi i n x} dx = \Gamma(s) \sigma^{-s} n^{-s}.$$

All termwise integrations are legitimate there because of the geometric factor q^n . Argument differentiation multiplies the negative Fourier mode by σn . For $\Re s > p+1$, this gives

$$J_p(s, z) := \int_0^1 \partial_x^p \zeta(1-s, x) R_z(x) dx = c\Gamma(s) \sigma^{p-s} \Phi_*(s-p, q). \quad (4.11)$$

This equation is then an identity of meromorphic continuations.

We now relate that continuation to the pointwise finite part; this is the step that cannot be replaced by substitution at $s = 0$. Write $r_j(z) = [x^j] R_z(x)$. For $p \geq 1$, the sole local term whose moving exponent crosses -1 at $s = 0$ is

$$r_p(z) A_p(s) x^{s-1}, \quad A_p(s) = \prod_{j=1}^p (s-j). \quad (4.12)$$

Indeed the singular part of $\partial_x^p \zeta(1-s, x)$ is $A_p(s)x^{s-p-1}$. The analytic part from $\zeta(1-s, x+1)$ has no pole in s after an argument derivative. Every other Taylor term of R_z has an exponent separated from -1 near zero. For such a term, finite-part integration commutes with coefficient extraction. The term (4.12) is different: its analytically continued integral is $r_p A_p(s)/s$, whereas the finite part of every coefficient $x^{-1} \log^j x$ on $(0, 1)$ is zero. It follows, with the entire amplitude retained, that

$$M_{m,p}(z) = m![s^m] \left(J_p(s, z) - \frac{r_p(z)A_p(s)}{s} \right), \quad p \geq 1. \quad (4.13)$$

One can make this argument on $(0, \delta)$ with a finite Taylor subtraction and a normally integrable remainder. The compensating integrals on $(\delta, 1)$ cancel the artificial splitting point. This also proves all holomorphy and coefficient interchanges used here.

Differentiating (4.10) at $x = 0$ gives

$$r_p(z) = \frac{c(-\sigma)^p}{p!} \Phi_*(-p, q), \quad p \geq 1. \quad (4.14)$$

Since $A_p(s) = (-1)^p p! E_p(s)$, substitution in (4.13) proves (4.7) for $p \geq 1$.

For $p = 0$, the product $\gamma_m R_z$ is integrable by (4.5). The pole in (4.4) is independent of x , so

$$\sum_{m \geq 0} M_{m,0}(z) \frac{s^m}{m!} = J_0(s, z) + \frac{1}{s} \int_0^1 R_z(x) dx = J_0(s, z) - \frac{1}{as}.$$

But $c\Phi_*(0, q) = 1/a$. This is exactly (4.7) with $E_0 = 1$. Thus the same formula accounts for both the global Hurwitz pole at $p = 0$ and the local moving endpoint singularity at $p \geq 1$. $\square$

4.3 Explicit examples and a check of the correction

At the first Stieltjes index the formula is

$$\text{FP}_x \int_0^1 \psi^{(p)}(x) R_z(x) dx = c\sigma^p \left\{ (\gamma + \text{Log } \sigma - H_p) \Phi_*(-p, q) - \Phi_*^{[1]}(-p, q) \right\}, \quad (4.15)$$

with $H_0 = 0$. The harmonic number is essential whenever $p > 0$. For $m = 1$ the complete formula is

$$\begin{aligned} M_{1,p}(z) = \frac{c\sigma^p}{2} \{ & \Phi_*^{[2]}(-p, q) - 2(\gamma + \text{Log } \sigma) \Phi_*^{[1]}(-p, q) \\ & + ((\gamma + \text{Log } \sigma)^2 + \zeta(2) - H_p^2 + H_p^{(2)}) \Phi_*(-p, q) \}. \end{aligned} \quad (4.16)$$

Here $H_0^{(2)} = 0$. The correction in this equation vanishes at $p = 0, 1$, as the general threshold predicts.

At $z = i\pi$, the normalized polylogarithm is identically one, and all positive order derivatives disappear. In particular,

$$\text{FP}_x \int_0^1 \frac{\psi'(x)}{\pi \cot(\pi x) - i\pi} dx = 1 - \gamma - \log(2\pi) + \frac{i\pi}{2}. \quad (4.17)$$

There is a direct independent verification of its constant term. At this z , $R'_z = e^{2\pi i x}$ and the finite boundary value of $[\psi R_z]_0^1$ is 1, because $\psi(x)R_z(x) \rightarrow -1$ as $x \downarrow 0$. Integration by parts therefore supplies precisely the 1 in (4.17). Omitting the local correction would miss it.

The local coefficients themselves require no special functions:

$$R'_z = 1 + 2zR_z + (z^2 + \pi^2)R_z^2, \quad R_z(0) = 0. \quad (4.18)$$

Thus $r_1 = 1$, $r_2 = z$, $r_3 = z^2 + \pi^2/3$, and $r_4 = z^3 + 2\pi^2 z/3$. In general r_p is a polynomial in z of degree $p - 1$. The difference between the raw spectral coefficient and the stated finite part is therefore a finite polynomial in z , not an additional unevaluated integral.

For a real parameter $b > 0$, taking the real and imaginary parts of the $p = m = 0$ formula at $z = ib$ gives ordinary real integrals. For example, with $q = (b - \pi)/(b + \pi)$,

$$\int_0^1 \frac{\gamma_0(x)K(x)}{K(x)^2 + b^2} dx = \frac{\pi}{2(b + \pi)}, \quad (4.19)$$

$$\int_0^1 \frac{\gamma_0(x)}{K(x)^2 + b^2} dx = \frac{\gamma + \log(2\pi) - (1 - q)\Phi_*^{[1]}(0, q)}{b(b + \pi)}. \quad (4.20)$$

At $b = 3\pi$, the last numerator becomes $\gamma + \log(2\pi) - \text{Li}_0^{[1]}(1/2)$. Equation (4.19) also follows directly from the digamma reflection formula, providing a separate check of the phase.

4.4 Repeated poles and the complete bounded-rational family

Corollary 4.2 (All repeated resolvents). *For every integer $r \geq 1$,*

$$\text{FP}_x \int_0^1 \frac{\gamma_m^{(p)}(x)}{(K(x) - z)^r} dx = \frac{1}{(r - 1)!} \partial_z^{r-1} M_{m,p}(z). \quad (4.21)$$

This is an ordinary absolutely convergent integral if $r > p$. Every rational function $Q(u)$ bounded at infinity and with no real pole admits a finite reduction of $\text{FP}_x \int_0^1 \gamma_m^{(p)}(x)Q(K(x)) dx$ to the right-hand sides of (4.21).

Proof. Parameter differentiation commutes with the finitely many local subtractions on compact subsets away from the real poles. Since $\partial_z^{r-1} R_z = (r - 1)! R_z^r$, the formula follows. Near zero the integrand has size $O(x^{r-p-1}(1 + |\log x|^m))$, proving the convergence assertion. Equivalently, the resonant coefficient $[x^p] R_z^r$ is zero when $r > p$, so all contact terms disappear from the differentiated formula.

Partial fractions express Q as a constant plus finitely many repeated resolvents. The constant contributes zero:

$$\text{FP}_x \int_0^1 \gamma_m^{(p)}(x) dx = 0.$$

For $p = 0$ this follows by integrating the Hurwitz spectral family and matching its constant and endpoint pole. For $p \geq 1$ it follows from the finite part of the endpoint difference of $\gamma_m^{(p-1)}$, using (4.5). The analytic parts at 0 and 1 agree, and the singular part has zero constant. This proves the complete finite reduction. $\square$

Repeated z derivatives introduce no new transcendental function class: $\partial_q \text{Li}_s(q) = \text{Li}_{s-1}(q)/q$, and $q(z)$ is rational. The reduction therefore stays within finitely many polylogarithmic order derivatives at explicit arguments. This proves the rational sector of the proposed nonpolynomial extension. It does not assert the same finite closure for arbitrary complex powers of $K - z$.

5 Continuous polygamma orders, Gamma primitives, and boundary values

5.1 An entire transform and its integer-order anomaly

We use the established generalized polygamma function of Espinosa and Moll [9], denoted here by $\Psi(v, x)$ to distinguish it from the ordinary digamma:

$$\Psi(v, x) = e^{-\gamma v} \partial_v \left(e^{\gamma v} \frac{\zeta(1+v, x)}{\Gamma(-v)} \right). \quad (5.1)$$

The removable values in this definition are understood. It is entire in v , satisfies $\partial_x \Psi(v, x) = \Psi(v+1, x)$, and equals $\psi^{(p)}(x)$ when $v = p \geq 0$. These are properties of the classical generalized function, not new definitions of polygamma.

Theorem 5.1 (Continuous-order resolvent and finite-part conversion). *For $\Re v < 1$ and $z \notin \mathbb{R}$, the following ordinary absolutely convergent integral holds:*

$$\mathcal{C}(v, z) := \int_0^1 \Psi(v, x) R_z(x) dx = c\sigma^v \left\{ (\gamma + \text{Log } \sigma) \Phi_*(-v, q) - \Phi_*^{[1]}(-v, q) \right\}. \quad (5.2)$$

The right side is entire in v . At a nonnegative integer p , its continuation and the pointwise finite part are related by

$$\text{FP}_x \int_0^1 \psi^{(p)}(x) R_z(x) dx = \mathcal{C}(p, z) - (-1)^p p! H_p r_p(z), \quad (5.3)$$

where the correction is zero for $p = 0$ without defining r_0 .

Proof. The singular part of (5.1) is a linear combination of x^{-v-1} and $x^{-v-1} \log x$ with entire coefficient functions of v . Multiplication by $R_z = O(x)$ gives locally uniform integrability when $\Re v < 1$. Thus the integral is holomorphic in that half-plane, including the removable point $v = 0$.

For $\Re v < 0$, divide (4.11) with $p = 0$ by $\Gamma(s)$ and set $s = -v$. Applying $e^{-\gamma v} \partial_v e^{\gamma v}$ yields (5.2). Holomorphic continuation gives the entire stated ordinary-integral strip. Finally, compare the right side at $v = p$ with (4.15), and use $c\sigma^p \Phi_*(-p, q) = (-1)^p p! r_p(z)$. This proves the exact conversion formula. $\square$

The last statement is a concrete instance of noncommutation between specializing a parameter and taking an endpoint finite part. It is not a defect of the generalized polygamma function. At $p = 1$ the correction in (5.3) is $+1$ for every nonreal z . At $p = 2$ it is $-3z$. These exact corrections are useful checks when a spectral formula is integrated into a symbolic system.

Integration by parts gives a compatible differential identity for the entire transform:

$$\mathcal{C}(v+1, z) = -2z\mathcal{C}(v, z) - (z^2 + \pi^2) \partial_z \mathcal{C}(v, z). \quad (5.4)$$

Initially take $\Re v < 0$, so the boundary term vanishes and $\int_0^1 \Psi(v, x) dx = 0$, and use (4.18). Both sides are entire in v , which proves the identity everywhere. For the pointwise finite-part moments $F_p(z) = \text{FP}_x \int \psi^{(p)} R_z$, the corresponding identity instead reads

$$F_{p+1}(z) = -2zF_p(z) - (z^2 + \pi^2) F'_p(z) + (-1)^p p! r_{p+1}(z). \quad (5.5)$$

The last term is exactly the finite part of $[\psi^{(p)} R_z]_0^1$. Thus the derivative hierarchy, the endpoint subtraction, and the polylogarithmic formula agree at every order.

5.2 Every balanced Gamma primitive

For $r \geq 0$, let

$$\mathcal{B}_r(x) = \frac{\zeta'(-r, x) + H_r \zeta(-r, x)}{r!}. \quad (5.6)$$

These are the standard balanced negapolygamma functions $\Psi(-r-1, x)$ [9]. Direct differentiation gives

$$\mathcal{B}_0(x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi), \quad \mathcal{B}'_r(x) = \mathcal{B}_{r-1}(x) \quad (r \geq 1).$$

The additive constants have therefore been specified completely.

Corollary 5.2 (The rational-kernel primitive ladder). *For every $r \geq 0$,*

$$\int_0^1 \mathcal{B}_r(x) R_z(x) dx = c\sigma^{-r-1} \left\{ (\gamma + \text{Log } \sigma) \Phi_*(r+1, q) - \Phi_*^{[1]}(r+1, q) \right\}. \quad (5.7)$$

All these integrals are ordinary convergent integrals.

This follows from (5.2) at $v = -r - 1$. There is also a useful direct check. Differentiate (4.11) in s at $s = r + 1$. The derivative of $\Gamma(s)$ supplies $H_r - \gamma$; the $H_r \zeta(-r, x)$ term in (5.6) cancels H_r exactly. That cancellation explains why the same coefficient $\gamma + \text{Log } \sigma$ works at every primitive level.

In particular,

$$\int_0^1 \log \Gamma(x) R_z(x) dx = -\frac{\log(2\pi)}{2a} + \frac{\Phi_*^{[1]}(1, q) - (\gamma + \text{Log } \sigma) \Phi_*(1, q)}{a^2}. \quad (5.8)$$

At the next level, the standard Barnes identity [11]

$$\zeta'(-1, x) = \zeta'(-1) - \log G(x) + (x-1) \log \Gamma(x)$$

shows that (5.7) is an explicit formula for the corresponding normalized Barnes- G combination, since $\zeta(-1, x) = -B_2(x)/2$. Here G is the standard Barnes function, with its usual asymptotic normalization, $G(x+1) = \Gamma(x)G(x)$, and $G(1) = 1$. The functional equation and the value at 1 alone would not exclude a multiplicative periodic factor. This supplies an exact primitive family relevant to Q9 in the source, without claiming to evaluate every unbalanced product containing G .

5.3 Boundary values and recovery of the Stieltjes functions

The resolvent also permits a useful inversion check. Given $u \in \mathbb{R}$, let $x_u \in (0, 1)$ be the unique point with $K(x_u) = u$. For $m \geq 0$ the ordinary moment $M_{m,0}$ has one-sided boundary values satisfying

$$M_{m,0}(u+i0) - M_{m,0}(u-i0) = \frac{2\pi i}{u^2 + \pi^2} \gamma_m(x_u). \quad (5.9)$$

Indeed the substitution $y = K(x)$ gives the Cauchy integral

$$M_{m,0}(z) = \int_{-\infty}^{\infty} \frac{\gamma_m(x_y)}{(y^2 + \pi^2)(y-z)} dy.$$

Its tail is integrable after the factor $(y-z)^{-1}$ is included, and its density is smooth locally at $y = u$. Splitting off that local value and integrating $1/(y-z)$ proves (5.9); this is the elementary Cauchy jump formula. The average of the two boundary values gives the corresponding principal-value integral at the interior pole.

On the boundary, $q = e^{-2\pi i x_u}$ in the upper half-plane and $q = e^{2\pi i x_u}$ in the lower one. Thus the two versions of (4.7) recover the classical Hurwitz–polylogarithm connection and all its Stieltjes order derivatives. This boundary interpretation is a consistency check and an exact inversion formula; it is not claimed as a new Hurwitz functional equation.

5.4 Verification and the precise new scope

The program `code/verify_resolvent.py` checks the Fourier and Riccati expressions for the local coefficients by exact rational function arithmetic, including both the Gamma polynomial and its harmonic correction. Its numerical checks compare the formulas with direct quadrature of digamma and generalized Stieltjes functions, local-subtraction quadrature for the finite parts, convergent integrals involving γ'_1 , and the first three balanced Gamma primitives. Complex-order tests use the defining Hurwitz formula (5.1). The numerical and exact records are distinct.

The all-index rational closure answers the specified rational sector of the source's nonpolynomial question. The local correction mechanism extends the repository's established finite-part calculus; it is not presented as an independently invented regularization. The continuous-order conversion and the primitive ladder explain which explicit boundary terms must accompany the resulting identities.

6 Nonpositive harmonic indices and a convergent Lerch generator

The incoming independent-order report reduces a depth-two word whose last index is a nonpositive integer. We extend that calculation to an arbitrary position and depth, prove a canonical finite reduction, and sum all zero-index decorations around one free index in a convergent Lerch generating function. The summation of powers by Bernoulli polynomials is classical; the contribution here is its compatibility with the entire relative harmonic interpolant at arbitrary endpoints, including its independent spectral derivatives. The interpolant itself is established in the incoming report, and its one-endpoint version is the interpolant of Ihara, Nakamura, and Yamamoto [8]. We do not claim literature-wide priority for these consequences.

6.1 Normalization and a local elimination theorem

For $\Re a, \Re b > 0$, use the normalization

$$\begin{aligned} Z_a(s_1, \dots, s_r) &= \sum_{n_1 > \dots > n_r \geq 0} \prod_{j=1}^r (n_j + a)^{-s_j}, \\ \mathcal{H}_{a,b}(w) &= Z_b(w) - \sum_{j=1}^{|w|} Z_a(w_{\leq j}) \mathcal{H}_{a,b}(w_{> j}), \quad \mathcal{H}_{a,b}(\emptyset) = 1. \end{aligned} \quad (6.1)$$

The Z_a are meromorphically continued from their convergence domain. The combined function $\mathcal{H}_{a,b}$ is entire in all spectral indices. For $a = b + N$, $N \geq 0$ an integer, it is the strict finite sum over $N > n_1 > \dots > n_r \geq 0$. In particular,

$$h_{a,b}(s) := \mathcal{H}_{a,b}(s) = \zeta(s, b) - \zeta(s, a), \quad h_{a,b}(1) = \psi(a) - \psi(b). \quad (6.2)$$

We take $B_n = B_n(0)$, so $B_1 = -1/2$, whereas $B_1(1) = 1/2$. This distinction is essential for strict inequalities.

Theorem 6.1 (Eliminating a fixed nonpositive index). *Let $m \geq 0$ and $q = m + 1$. All other indices below are arbitrary independent complex numbers. With unchanged prefix and suffix words suppressed, an interior index obeys*

$$\begin{aligned} \mathcal{H}_{a,b}(\dots, s, -m, t, \dots) &= \frac{1}{q} \sum_{k=0}^q \binom{q}{k} \{ B_{q-k} \mathcal{H}_{a,b}(\dots, s - k, t, \dots) \\ &\quad - B_{q-k}(1) \mathcal{H}_{a,b}(\dots, s, t - k, \dots) \}. \end{aligned} \quad (6.3)$$

At the two boundaries the formulas are

$$\mathcal{H}_{a,b}(-m, t, \dots) = \frac{1}{q} \left\{ B_q(a) \mathcal{H}_{a,b}(t, \dots) - \sum_{k=0}^q \binom{q}{k} B_{q-k}(1) \mathcal{H}_{a,b}(t-k, \dots) \right\}, \quad (6.4)$$

$$\mathcal{H}_{a,b}(\dots, s, -m) = \frac{1}{q} \left\{ \sum_{k=0}^q \binom{q}{k} B_{q-k} \mathcal{H}_{a,b}(\dots, s-k) - B_q(b) \mathcal{H}_{a,b}(\dots, s) \right\}. \quad (6.5)$$

The singleton rule is

$$h_{a,b}(-m) = \frac{B_q(a) - B_q(b)}{q}. \quad (6.6)$$

These are identities of entire functions in every remaining index.

Proof. The classical finite power-sum identity [6] gives

$$\sum_{z < x < y} x^m = \frac{B_q(y) - B_q(z+1)}{q}, \quad (6.7)$$

when the variables belong to a common unit lattice and the sum has the indicated strict endpoints. The upper and lower boundary variants replace y by a and $z+1$ by b , respectively. This explains the formulas for finite sums, but agreement at integer endpoint differences alone would not justify their analytic continuation. We instead prove the identities directly in (6.1).

For Z_a , summing an interior or final nonpositive index by (6.7) proves (6.3) and (6.5), with $\mathcal{H}_{a,b}$ replaced by Z_a and the final lower endpoint b replaced by a . One first chooses the unfixed indices in a domain of absolute convergence and then continues meromorphically. For a first index use

$$\sum_{n_1 > n_2} (n_1 + a)^{-z} = \zeta(z, n_2 + a + 1), \quad \zeta(-m, x + 1) = -\frac{B_q(x + 1)}{q}.$$

Taking the remaining indices sufficiently far into their convergence domain justifies continuation of z to $-m$ inside this expression. Consequently the top-index identity for Z_a is

$$Z_a(-m, t, \dots) = -\frac{1}{q} \sum_{k=0}^q \binom{q}{k} B_{q-k}(1) Z_a(t-k, \dots). \quad (6.8)$$

All these statements concern generic values on the indicated spectral hyperplane, followed by meromorphic continuation; they assign no directional value to an individual singular Z_a at an intersection.

Now induct on the depth in (6.1). For the top formula substitute (6.8) into every prefix of length at least two. The prefix of length one contributes $-Z_a(-m) \mathcal{H}_{a,b}(t, \dots) = B_q(a) \mathcal{H}_{a,b}(t, \dots)/q$. Each remaining group is exactly a shorter convolution identity, giving (6.4). The bottom formula follows in the same way. Its two exceptional terms are the last prefix and the singleton suffix; the factors involving $B_q(a)$ cancel, leaving $B_q(b)$.

For clarity, consider an interior letter and write its nonempty words on the two sides as L and R . In the deconcatenation sum, cuts strictly before L 's last letter use the induction hypothesis on the suffix; cuts after the first letter of R use the interior Z_a identity on the prefix. The two adjacent cuts are

$$Z_a(L) \mathcal{H}_{a,b}(-m, R), \quad Z_a(L, -m) \mathcal{H}_{a,b}(R).$$

After the top and bottom substitutions their exceptional terms are $B_q(a) Z_a(L) \mathcal{H}_{a,b}(R)/q$ and its negative. They cancel. The remaining left-shifted and right-shifted terms are precisely the recursions for the two shorter words in (6.3). This proves that formula on generic remaining indices. Entireness of $\mathcal{H}_{a,b}$ then extends all the identities to every remaining index, including coincident resonances. $\square$

For $m = 0$, the interior rule takes the particularly transparent form

$$\mathcal{H}_{a,b}(\dots, s, 0, t, \dots) = \mathcal{H}_{a,b}(\dots, s-1, t, \dots) - \mathcal{H}_{a,b}(\dots, s, t-1, \dots) - \mathcal{H}_{a,b}(\dots, s, t, \dots). \quad (6.9)$$

The last term records the strict gap's missing endpoint. Changing $B_1(1)$ to B_1 incorrectly deletes it.

6.2 A canonical polynomial normal form

Suppose a word contains k free indices, in their original order $s_1, \dots, s_k$, and fixed indices $-m_1, \dots, -m_e$, where $m_j \geq 0$. Set $M = \sum_{j=1}^e (m_j + 1)$.

Theorem 6.2 (Finite reduction at arbitrary depth). *There are explicitly computable polynomials $C_\nu(a, b) \in \mathbb{Q}[a, b]$ such that*

$$\boxed{\mathcal{H}_{a,b}(w) = \sum_{\substack{\nu \in \mathbb{N}_0^k \\ |\nu| \leq M}} C_\nu(a, b) \mathcal{H}_{a,b}(s_1 - \nu_1, \dots, s_k - \nu_k), \quad \deg C_\nu \leq M - |\nu|.} \quad (6.10)$$

For $k = 0$ the result is a polynomial multiplying the empty word. Every elimination order in Theorem 6.1 gives the same coefficient polynomials when the free indices are kept formally distinct. The bound concerns this explicitly specified finite normal form and does not assert a minimal arithmetic depth.

Proof. Every local rule deletes one fixed index. A deleted index $-m$ has budget $q = m + 1$: it either produces an endpoint polynomial of degree at most q , or shifts an adjacent index by $k \leq q$. If the adjacent index is also fixed, its later budget increases by k ; if it is free, its final shift increases by k . Induction therefore proves termination and the degree bound in (6.10).

Here is a canonical description of the coefficients. For a fixed nonpositive block v , let

$$\Pi_v(A, B) = \mathcal{H}_{A,B}(v), \quad \Pi_\emptyset(A, B) = 1.$$

The first part proves that these are polynomials. Write $w = (v_0, s_1, v_1, \dots, s_k, v_k)$ and expand the polynomial

$$G_w(\mathbf{y}; a, b) = \Pi_{v_0}(a, y_1 + 1) \prod_{j=1}^{k-1} \Pi_{v_j}(y_j, y_{j+1} + 1) \Pi_{v_k}(y_k, b) = \sum_{\nu} C_\nu(a, b) \mathbf{y}^\nu. \quad (6.11)$$

For integer endpoint differences, fix the free summation variables $y_1 > \dots > y_k$. The sums in the intervening gaps are independent, and their product is exactly (6.11).

To identify these coefficients with those of any already proved local reduction, compare the two finite sums for all independent s_j . Distinct ordered tuples of positive lattice points give linearly independent functions $\prod_j y_j^{-s_j}$: this is elementary finite linear independence of exponential functions, applied successively in the s_j . Thus the difference of the two coefficient polynomials vanishes at every lattice configuration $a = b + N$, $y_j = b + n_j$, $N > n_1 > \dots > n_k \geq 0$, with $b > 0$. These configurations are Zariski dense. Indeed a polynomial vanishing on all integer tuples satisfying strict inequalities vanishes by successive univariate polynomial uniqueness, and b ranges over an interval. Hence the coefficient polynomials agree identically. This is polynomial uniqueness after the analytic identities have already been proved; it is not uniqueness of a holomorphic interpolation from integer endpoint values. $\square$

In the one-free-index case, every depth reduces to the finite sum

$$\mathcal{H}_{a,b}(v_0, s, v_1) = \sum_{\nu=0}^M C_\nu(a, b) h_{a,b}(s - \nu). \quad (6.12)$$

All derivatives in the free index commute with this identity. At $s - \nu = 1$ use the finite value

$$h_{a,b}^{(r)}(1) = (-1)^r \{\gamma_r(b) - \gamma_r(a)\}, \quad h'_{a,b}(0) = \log \Gamma(b) - \log \Gamma(a). \quad (6.13)$$

At positive integer arguments > 1 , the undifferentiated differences are polygamma differences. These standard Hurwitz specializations are recorded in [6]. Differentiation of a *fixed* index $-m$ is a different problem and is not licensed by a formula proved only on that spectral hyperplane.

Two concise examples are

$$\mathcal{H}_{a,b}(0, s, 0) = -h_{a,b}(s-2) + (a+b-1)h_{a,b}(s-1) - b(a-1)h_{a,b}(s), \quad (6.14)$$

$$\begin{aligned} \mathcal{H}_{a,b}(-1, s, -1) &= \frac{1}{4} \{-h_{a,b}(s-4) + (A+B+1)h_{a,b}(s-2) \\ &\quad + (B-A)h_{a,b}(s-1) - ABh_{a,b}(s)\}. \end{aligned} \quad (6.15)$$

Here $A = a(a-1)$ and $B = b(b-1)$ in the second identity. The absent shift $s-3$ in the second identity is an exact cancellation. It follows by expanding $(A-y^2-y)(y^2-y-B)/4$. For example, its derivative at $s=2$ is

$$\begin{aligned} \partial_s \mathcal{H}_{a,b}(-1, s, -1)|_{s=2} &= \frac{1}{4} \{-\zeta'(-2, b) + \zeta'(-2, a) \\ &\quad + (A+B+1) \log \frac{\Gamma(b)}{\Gamma(a)} + (B-A)\{\gamma_1(a) - \gamma_1(b)\} \\ &\quad - AB\{\zeta'(2, b) - \zeta'(2, a)\}\}. \end{aligned} \quad (6.16)$$

6.3 Summing arbitrarily many zero indices

For a block of p zero indices, polynomial continuation of the finite count gives

$$\Pi_{\{0\}^p}(A, B) = \binom{A-B}{p}. \quad (6.17)$$

Consequently (6.11) for a single free index uses the polynomial $\binom{a-y-1}{p} \binom{y-b}{q}$. All such polynomials can be summed in one analytic formula.

Theorem 6.3 (Lerch generator for zero decorations). *Let $\Re a, \Re b > 0$. There is a function $\mathcal{L}_{a,b}(s; t)$ entire in s and holomorphic for $|t| < 2\pi$ such that, when $\Re t < 0$,*

$$\mathcal{L}_{a,b}(s; t) = e^{bt} \Phi(e^t, s, b) - e^{at} \Phi(e^t, s, a). \quad (6.18)$$

Here $\Phi(z, s, c) = \sum_{n \geq 0} z^n (n+c)^{-s}$ for $|z| < 1$, with principal powers of $n+c$. The combination in (6.18), not its separate summands, is continued through $t=0$. It satisfies

$$\mathcal{L}_{a,b}(s; 0) = h_{a,b}(s), \quad \partial_t \mathcal{L}_{a,b}(s; t) = \mathcal{L}_{a,b}(s-1; t), \quad \mathcal{L}_{a,b}(s; t) = \sum_{n \geq 0} h_{a,b}(s-n) \frac{t^n}{n!}. \quad (6.19)$$

The last series converges locally normally for $|t| < 2\pi$, including all fixed spectral derivatives. For $|u|, |v| < 1/2$ one has the convergent identity

$$\boxed{\sum_{p,q \geq 0} \mathcal{H}_{a,b}(\{0\}^p, s, \{0\}^q) u^p v^q = (1+u)^{a-1} (1+v)^{-b} \mathcal{L}_{a,b}(s; \log(1+v) - \log(1+u))}. \quad (6.20)$$

Powers and logarithms on the right are their germs at $u = v = 0$.

Proof. Define

$$K_{a,b}(y) = \frac{e^{-by} - e^{-ay}}{1 - e^{-y}}. \quad (6.21)$$

The apparent pole at $y = 0$ is removable, with value $a - b$. Its only other possible finite poles are $2\pi i\mathbb{Z} \setminus \{0\}$. Thus for $|t| < 2\pi$ the function $K_{a,b}(x - t)$ is analytic near every $x \geq 0$ and decreases exponentially as $x \rightarrow +\infty$, locally uniformly in t, a, b . Initially for $\Re s > 0$ put

$$\mathcal{L}_{a,b}(s; t) = \frac{1}{\Gamma(s)} \int_0^\infty x^{s-1} K_{a,b}(x - t) dx. \quad (6.22)$$

For any $N \geq 1$, subtract the Taylor polynomial of $K_{a,b}(x - t)$ at $x = 0$ in the integral from 0 to 1, and add

$$\frac{1}{\Gamma(s)} \sum_{j=0}^{N-1} \frac{K_{a,b}^{(j)}(-t)}{j!(s+j)}.$$

The subtracted integral is holomorphic for $\Re s > -N$. The displayed possible poles are cancelled by the simple zeros of $1/\Gamma(s)$. Since N is arbitrary, this proves joint entireness in s and holomorphy for $|t| < 2\pi$, with locally normal spectral derivatives.

For $\Re t < 0$, geometric expansion in (6.22) and the Gamma integral give (6.18), first for $\Re s > 0$ and then for all s . At $t = 0$ the ordinary Hurwitz integral gives $h_{a,b}(s)$, initially for $\Re s > 1$, and then by continuation. Differentiating the absolutely convergent Lerch series for $\Re t < 0$ proves the order-lowering identity in (6.19); continuation and Taylor's theorem give the remaining assertions.

The right side of (6.20) is holomorphic for $|u|, |v| < 1/2$, because the absolute value of the difference of its logarithms is less than $2 \log 2 < 2\pi$. To identify a coefficient, formally replace $h_{a,b}(s - n)$ in (6.19) by y^n . Its right side becomes e^{yt} . Including the prefactors in (6.20) gives

$$(1 + u)^{a-y-1} (1 + v)^{y-b}.$$

Its coefficient of $u^p v^q$ is precisely $\binom{a-y-1}{p} \binom{y-b}{q}$. At each fixed bidegree this is a finite polynomial calculation. The canonical gap formula then identifies it with $\mathcal{H}_{a,b}(\{0\}^p, s, \{0\}^q)$, proving the identity and the convergence of its left side. $\square$

The branch cancellation can also be read as a Gamma separation: $\Gamma(s)\mathcal{L}_{a,b}(s; t)$ is the ordinary Mellin transform of the explicit kernel (6.21); its apparent spectral poles at nonpositive integers are entirely accounted for by $\Gamma(s)$. This proves a generating formula for the specified one-free-order family. It does not settle a minimality question for several arbitrary independent nonzero indices.

Setting $u = v$ gives the exact insertion identity

$$\sum_{p+q=r} \mathcal{H}_{a,b}(\{0\}^p, s, \{0\}^q) = \binom{a-b-1}{r} h_{a,b}(s). \quad (6.23)$$

The generating function also has normalized antiderivatives:

$$\int_0^t \mathcal{L}_{a,b}(s; \tau) d\tau = \mathcal{L}_{a,b}(s+1; t) - h_{a,b}(s+1). \quad (6.24)$$

At a nonpositive order,

$$\mathcal{L}_{a,b}(-m; t) = \partial_t^m \frac{e^{bt} - e^{at}}{1 - e^t}, \quad m \geq 0, \quad (6.25)$$

where $t = 0$ is removable. At $a = b + N$, $N \geq 0$ integer, $\mathcal{L}_{a,b}(s; t) = \sum_{n=0}^{N-1} (n+b)^{-s} e^{(n+b)t}$; the double generating function becomes a polynomial of total degree at most $N - 1$ when $N \geq 1$ and is zero when $N = 0$.

6.4 Polynomial endpoint primitives and Stieltjes moments

The polynomial Hurwitz integration mechanism used here is classical: Espinosa and Moll give a general integration-by-parts formula and its polynomial specialization [10, Theorem 2.1 and Example 2.2]. The canonical manuscript [1], in the Moments section of `chapters/07-integration.tex`, already states the unit-interval Hurwitz transform, gives the first Stieltjes moment, and identifies the complete homogeneous coefficients for every Stieltjes order. We state the fully indexed arbitrary-endpoint version, with the normalization required by (6.12). This is an explicit extension and restatement of the existing moment mechanism, and is not counted as a new general Stieltjes-moment principle.

For $d \geq 0$, write $\Delta = a - b$ and define

$$P_{d;a,b}(s) = \int_b^a (x - b)^d h_{x,b}(s) dx. \quad (6.26)$$

Paths lie in the right half-plane. The integral is path independent and entire in s . Initially away from singularities of the individual displayed terms,

$$\begin{aligned} P_{d;a,b}(s) &= \frac{\Delta^{d+1}}{d+1} \zeta(s, b) \\ &+ \sum_{j=0}^d \frac{d!}{(d-j)!} \frac{\Delta^{d-j} \zeta(s-j-1, a) - \mathbf{1}_{j=d} \zeta(s-j-1, b)}{(s-1)(s-2) \cdots (s-j-1)}. \end{aligned} \quad (6.27)$$

This follows by repeated integration by parts using $\partial_x \zeta(s-1, x) = -(s-1)\zeta(s, x)$. The definite integral proves that every apparent pole in the combined right side is removable. Expanding a coefficient polynomial $C_\nu(x, b)$ in powers of $x - b$ and using (6.27) therefore integrates every one-free-order word and every one of its free-order derivatives in a finite Hurwitz basis, with the constant fixed at $a = b$.

To state all resonance coefficients, put

$$c_{j,n} = [z^n] \prod_{\ell=1}^j (1 - z/\ell)^{-1}, \quad c_{0,n} = \mathbf{1}_{n=0}. \quad (6.28)$$

These are finite generalized harmonic expressions:

$$\sum_{n \geq 0} c_{j,n} z^n = \exp \left(\sum_{r \geq 1} \frac{H_j^{(r)}}{r} z^r \right), \quad H_j^{(r)} = \sum_{\ell=1}^j \ell^{-r}.$$

For example $c_{j,0} = 1$, $c_{j,1} = H_j$ and $c_{j,2} = (H_j^2 + H_j^{(2)})/2$. Equivalently, $c_{j,n}$ is the complete exponential Bell polynomial in $H_j, 1!H_j^{(2)}, \dots, (n-1)!H_j^{(n)}$, divided by $n!$.

Corollary 6.4 (Explicit arbitrary-endpoint form of the moment calculus). *For $d, m \geq 0$ and $\Re a, \Re b > 0$,*

$$\begin{aligned} \int_b^a (x - b)^d \gamma_m(x) dx &= (-1)^{m+1} m! \sum_{j=0}^d (-1)^j \binom{d}{j} \sum_{r=0}^{m+1} \frac{c_{j,m+1-r}}{r!} \\ &\cdot \left\{ \Delta^{d-j} \zeta^{(r)}(-j, a) - \mathbf{1}_{j=d} \zeta^{(r)}(-j, b) \right\}. \end{aligned} \quad (6.29)$$

Here derivatives of ζ are with respect to its first argument.

Proof. Set $s = 1 + z$ in (6.27). The denominator in its j th term has expansion

$$\frac{1}{z(z-1) \cdots (z-j)} = \frac{(-1)^j}{j! z} \sum_{n \geq 0} c_{j,n} z^n.$$

Use $\zeta(1+z, b) = z^{-1} + \sum_{m \geq 0} (-1)^m \gamma_m(b) z^m / m!$ and the ordinary Taylor expansion of each $\zeta(z-j, a)$. The coefficient of z^{-1} cancels, as it must by entireness of $P_{d,a,b}$. The coefficient of z^m , multiplied by $m!$, equals

$$(-1)^m \frac{\Delta^{d+1}}{d+1} \gamma_m(b) + m! \sum_{j=0}^d (-1)^j \binom{d}{j} \sum_{r=0}^{m+1} \frac{c_{j,m+1-r}}{r!} \left\{ \Delta^{d-j} \zeta^{(r)}(-j, a) - \mathbf{1}_{j=d} \zeta^{(r)}(-j, b) \right\}.$$

On the integral side, (6.13) gives $(-1)^m \{ \Delta^{d+1} \gamma_m(b) / (d+1) - \int_b^a (x-b)^d \gamma_m(x) dx \}$. Cancel the common endpoint term and solve for the integral. All coefficient extractions are locally normal on compact endpoint sets in the right half-plane. $\square$

For $d = 0$ this recovers the familiar normalized primitive

$$\int_b^a \gamma_m(x) dx = \frac{(-1)^{m+1}}{m+1} \{ \zeta^{(m+1)}(0, a) - \zeta^{(m+1)}(0, b) \}. \quad (6.30)$$

For $d = 1$, since $c_{1,n} = 1$ for every n ,

$$\int_b^a (x-b) \gamma_m(x) dx = (-1)^{m+1} m! \left\{ \frac{\Delta \zeta^{(m+1)}(0, a)}{(m+1)!} - \sum_{r=0}^{m+1} \frac{\zeta^{(r)}(-1, a) - \zeta^{(r)}(-1, b)}{r!} \right\}. \quad (6.31)$$

In particular $m = 0$ reduces to an integral of the digamma function, and Lerch's identity converts the $\zeta'(0, a)$ term to $\log \Gamma(a) - \frac{1}{2} \log(2\pi)$.

6.5 Exact scope and further questions

The identities above add arbitrary-depth finite reductions and a convergent all-zero-decoration generator to the previously established depth-two calculus. They do not evaluate nonsymmetric positive-index coordinates such as the incoming harmonic Stieltjes parameter η . No statement about the unresolved S_6 or S_8 reductions is made.

Natural next questions are the following.

1. Construct and prove a jointly convergent analogue of (6.20) with two or more independently moving free indices and arbitrary zero blocks between them. The finite coefficient polynomials (6.11) are already explicit; the remaining task is a useful analytic kernel and a precise simultaneous branch domain.
2. Differentiate an eliminated index before specializing it to $-m$. Determine a finite set of additional ordered coordinates needed at the first such derivative. The present identities only differentiate the free indices, so this question asks for a real extension.
3. For rational endpoints, combine (6.29) with finite character projections to give algorithms for polynomial moments in Dirichlet L -derivatives, retaining every conductor logarithm.
4. Formalize the Bernoulli deletion rules, the polynomial gap normal form, and the finite coefficient extraction. Their analytic interface is the already established entire interpolant and the explicit normalized Mellin continuation in (6.22).

7 The complete quartic Tornheim layer and the formal jets at every order

The incoming *Independent Orders and Exact Reductions* report reduces every cubic directional derivative of the Tornheim function to two scalar coordinates and asks for its quartic layer [3].

We determine that layer explicitly. Three coordinates suffice: the diagonal fourth derivative, one convergent logarithmic Gamma moment, and one absolutely convergent integral of ordinary polylogarithmic order derivatives. We also determine the exact dimension of the homogeneous jet space allowed by the source's symmetry, axis, and Euler-plane identities. These are statements about a specified formal relation space, not about arithmetic independence of the constants.

7.1 The analytic normalization and its exact restrictions

Write

$$T(A, B, C) = \sum_{m, n \geq 1} m^{-A} n^{-B} (m+n)^{-C}, \quad L = \log(2\pi), \quad z_j = \zeta^{(j)}(0), \quad y_j = \zeta^{(j)}(-1).$$

The series is initially used only in its domain of absolute convergence; elsewhere T denotes its meromorphic continuation. The origin is not a joint regular point. For a direction (a, b, c) satisfying $(a+c)(b+c) \neq 0$, put

$$W_{a,b,c}(t) = T(at, bt, ct),$$

where the restriction is formed before continuation to $t = 0$.

The source constructs the unique symmetric holomorphic germ $R(A, B, C) = R(B, A, C)$ such that

$$T(A, B, C) = \zeta(A)\zeta(B) - \frac{C\zeta(B-1)}{A+C} - \frac{C\zeta(A-1)}{B+C} + CR(A, B, C). \quad (7.1)$$

For completeness, Jonquière's expansion [6, 25.12.12] gives this by subtracting the following six local terms from the Mellin kernel:

$$\begin{aligned} \mathcal{S}(A, B; x) = & \Gamma(1-A)\Gamma(1-B)x^{A+B-2} + \Gamma(1-A)\zeta(B)x^{A-1} + \Gamma(1-B)\zeta(A)x^{B-1} \\ & - \Gamma(1-A)\zeta(B-1)x^A - \Gamma(1-B)\zeta(A-1)x^B + \zeta(A)\zeta(B). \end{aligned} \quad (7.2)$$

The subtracted remainder and every fixed parameter derivative are normally integrable after multiplication by x^{C-1} in a sufficiently small common polydisc. Before removing the Gamma factors the polar terms contain $\Gamma(1-A)/\Gamma(1+C)$. Their difference from 1 is divisible by $A+C$, since it vanishes when $C = -A$. This proves that the difference between the resulting polar model and (7.1), divided by C , is holomorphic.

We will use precisely two functional restrictions of this germ:

$$CR(0, 0, C) = \zeta(C-1) - \zeta(C) - \frac{5}{12}, \quad (7.3)$$

$$\begin{aligned} CR(A, 0, C) + AR(C, 0, A) = & \zeta(A)\zeta(C) - \zeta(A+C) + \frac{\zeta(A) + \zeta(C)}{2} \\ & + \zeta(-1) + \zeta(A-1) + \zeta(C-1). \end{aligned} \quad (7.4)$$

The first follows by counting pairs with prescribed sum. The second follows from the Euler stuffle formula for $T(A, 0, C) + T(C, 0, A)$. Both are holomorphic identities near the origin. They were proved in the incoming report; their use here does not constitute a new proof of the cubic theorem.

7.2 All quartic directions in three explicit coordinates

All partial derivatives of R below are taken at the origin. Set

$$\Lambda = W_{1,1,1}^{(4)}(0), \quad U = R_{AAA}, \quad \Xi = R_{AAB}.$$

The notation Ξ distinguishes this new coordinate from the several unrelated uses of κ in the source articles.

Lemma 7.1 (The degree-three remainder constraints). *Write*

$$\begin{aligned} u &= R_{AAA} = R_{BBB}, & v &= R_{AAB} = R_{ABB}, & w &= R_{AAC} = R_{BBC}, \\ x &= R_{ABC}, & y &= R_{ACC} = R_{BCC}, & z &= R_{CCC}. \end{aligned}$$

Then

$$z = \frac{y_4 - z_4}{4}, \quad w = \frac{z_2^2 - z_4}{4}, \quad u + 3y = -\frac{L}{2}z_3 - z_4, \quad (7.5)$$

$$v + x = \frac{\Lambda + 8Lz_3 - 12z_2^2 + 16z_4}{24}. \quad (7.6)$$

Proof. The coefficient of C^4 in (7.3) is $z/6$, and its right-hand side is $(y_4 - z_4)/24$. In (7.4), the coefficients of A^2C^2 and A^3C are respectively w and $u/6 + y/2$. The corresponding right-hand coefficients are $(z_2^2 - z_4)/4$ and $(z_1z_3 - z_4)/6$, where $z_1 = -L/2$. This proves (7.5).

On the diagonal, (7.1) becomes

$$T(t, t, t) = \zeta(t)^2 - \zeta(t-1) + tR(t, t, t).$$

Its fourth derivative is

$$\Lambda = -z_4 - 4Lz_3 + 6z_2^2 - y_4 + 8u + 24v + 24w + 24x + 24y + 4z.$$

Substituting (7.5) leaves $\Lambda = -8Lz_3 + 12z_2^2 - 16z_4 + 24(v + x)$, as required. $\square$

Theorem 7.2 (Every quartic Tornheim ray). *For $(a + c)(b + c) \neq 0$, the restricted function $W_{a,b,c}$ is holomorphic at zero and*

$$\begin{aligned} W_{a,b,c}^{(4)}(0) &= abc^2\Lambda + 4c(a^3 + b^3 - c^2(a + b))U + 12abc(a + b - 2c)\Xi \\ &\quad - 2(ab(a^2 + b^2 - 4c^2) + c^3(a + b))Lz_3 \\ &\quad + 3(2a^2b^2 + c^2(a^2 + b^2 - 4ab))z_2^2 \\ &\quad + \left(-\frac{a^4 + b^4}{2} + 16abc^2 - 3c^2(a^2 + b^2) - 4c^3(a + b) - c^4 \right) z_4 \\ &\quad + \left(c^4 - \frac{cb^4}{a + c} - \frac{ca^4}{b + c} \right) y_4. \end{aligned} \quad (7.7)$$

The scalar U is given by the convergent Gamma series (7.10); Ξ is given by the convergent integral (7.14). No unevaluated partial derivative of T appears in these two coordinates.

Proof. On an admissible ray the two rational polar terms in (7.1) become constant multiples of holomorphic one-variable zeta functions, proving the asserted holomorphy. The degree-three homogeneous part of R is

$$\frac{u}{6}(A^3 + B^3) + \frac{v}{2}AB(A + B) + \frac{w}{2}C(A^2 + B^2) + xABC + \frac{y}{2}C^2(A + B) + \frac{z}{6}C^3.$$

Therefore the fourth derivative of (7.1) is

$$\begin{aligned} & -\frac{a^4 + b^4}{2}z_4 - 2ab(a^2 + b^2)Lz_3 + 6a^2b^2z_2^2 - c\left(\frac{b^4}{a + c} + \frac{a^4}{b + c}\right)y_4 \\ & + 4c\left(u(a^3 + b^3) + 3vab(a + b) + 3wc(a^2 + b^2) + 6xabc + 3yc^2(a + b) + zc^3\right). \end{aligned}$$

Put $u = U, v = \Xi$ and substitute (7.5)–(7.6). Collecting the five ordinary zeta combinations and the three retained coordinates gives (7.7). $\square$

A quartic cyclic identity with no additional coordinates. Let $a + b, a + c, b + c$ all be nonzero and set $e_1 = a + b + c$, $e_2 = ab + bc + ca$, $e_3 = abc$. Then

$$\begin{aligned} \sum_{\text{cyc}} W_{a,b,c}^{(4)}(0) &= e_1 e_3 \Lambda + (-4e_1^2 e_2 + 8e_2^2 + 12e_1 e_3) L z_3 \\ &\quad + (12e_2^2 - 36e_1 e_3) z_2^2 \\ &\quad + (-2e_1^4 + 4e_1^2 e_2 - 2e_2^2 + 24e_1 e_3) z_4. \end{aligned} \quad (7.8)$$

Indeed, the cyclic coefficients of U and Ξ in (7.7) vanish identically. Pairing the two terms with each denominator cancels every y_4 term. The remaining polynomial coefficients give the displayed formula. For example,

$$W_{1,2,3}^{(4)}(0) + W_{2,3,1}^{(4)}(0) + W_{3,1,2}^{(4)}(0) - 36\Lambda = -184Lz_3 + 156z_2^2 - 386z_4. \quad (7.9)$$

The Euler diagonal supplies an independent elementary check:

$$W_{1,0,1}^{(4)}(0) = -2Lz_3 + 3z_2^2 - \frac{17}{2}z_4,$$

which follows directly from $T(t, 0, t) = (\zeta(t)^2 - \zeta(2t))/2$.

More generally, the exact coordinate map

$$(a, b, c) \mapsto (abc^2, 4c(a^3 + b^3 - c^2(a + b)), 12abc(a + b - 2c))$$

gives a sufficient algebraic criterion for finite linear combinations of quartic rays to reduce entirely to the ordinary zeta derivatives displayed in (7.7). This criterion is exact in the stated three-coordinate normal form. Additional arithmetic relations among the coordinates could yield further reductions.

7.3 Convergent representatives for the two additional coordinates

Define the Stirling remainder

$$\rho(n) = \log \Gamma(n + 1) - \left(n + \frac{1}{2}\right) \log n + n - \frac{L}{2} - \frac{1}{12n}.$$

The source's boundary-germ theorem states, for every $j \geq 0$,

$$\partial_A^j R(0, 0, 0) = (-1)^j \sum_{n \geq 1} (\log n)^j \rho(n) - y_{j+1} - \frac{1}{2} z_{j+1} - y_j + \frac{(-1)^j \gamma_j}{12}.$$

Consequently the coordinate needed here is

$$U = - \sum_{n \geq 1} (\log n)^3 \rho(n) - y_4 - \frac{1}{2} z_4 - y_3 - \frac{\gamma_3}{12}. \quad (7.10)$$

This is a specialization of an existing result, not a separate novel Gamma-moment identity. Absolute convergence follows from $\rho(n) = O(n^{-3})$.

For the genuinely mixed coordinate let

$$f_j(x) = \partial_s^j \text{Li}_s(e^{-x}) \Big|_{s=0}, \quad h(x) = \log x + \gamma, \quad j = 1, 2.$$

For fixed $x > 0$, the defining series are absolutely convergent:

$$f_1(x) = - \sum_{n \geq 1} (\log n) e^{-nx}, \quad f_2(x) = \sum_{n \geq 1} (\log n)^2 e^{-nx}.$$

Put

$$P(x) = \frac{h(x)^3 + \zeta(2)h(x)}{x^2} + \frac{z_1(h(x)^2 + \zeta(2)) + z_2h(x)}{x} - y_1(h(x)^2 + \zeta(2)) - y_2h(x) + z_1z_2, \quad (7.11)$$

and define the ordinary absolutely convergent integral

$$I = \int_0^1 \frac{f_2(x)f_1(x) - P(x)}{x} dx + \int_1^\infty \frac{f_2(x)f_1(x)}{x} dx. \quad (7.12)$$

Proposition 7.3 (Explicit mixed polylogarithmic moment). *Let $g = \gamma$, $h_2 = \zeta(2)$, $h_3 = \zeta(3)$, and set*

$$E = -\frac{g^3 + gh_2}{2} - \frac{3g^2 + h_2}{4} - \frac{3g}{4} - \frac{3}{8} - (g^2 + h_2 + 2g + 2)z_1 - (g + 1)z_2 + gz_1z_2 - \frac{g^3 + 3gh_2 + 2h_3}{3}y_1 - \frac{g^2 + h_2}{2}y_2. \quad (7.13)$$

Then

$$\Xi = I + E. \quad (7.14)$$

Proof. Define the normally convergent subtracted Mellin integral

$$I(A, B) = \int_0^1 \frac{\text{Li}_A(e^{-x}) \text{Li}_B(e^{-x}) - \mathcal{S}(A, B; x)}{x} dx + \int_1^\infty \frac{\text{Li}_A(e^{-x}) \text{Li}_B(e^{-x})}{x} dx.$$

The $C = 0$ specialization of the proof of (7.1) gives the holomorphic identity

$$R(A, B, 0) = I(A, B) + \frac{\Gamma(1-A)\Gamma(1-B)}{A+B-2} + \frac{\Gamma(1-A)\zeta(B)}{A-1} + \frac{\Gamma(1-B)\zeta(A)}{B-1} + \gamma\zeta(A)\zeta(B) - \frac{\Gamma(1-A)-1}{A}\zeta(B-1) - \frac{\Gamma(1-B)-1}{B}\zeta(A-1). \quad (7.15)$$

Both apparent quotients at $A = 0$ or $B = 0$ are removable. Differentiating $\mathcal{S}$ twice in A and once in B at zero produces precisely $P(x)$. Thus $I_{AAB}(0, 0) = I$.

To see convergence directly, write the order-differentiated Jonquière expansion as

$$f_j(x) = \frac{g_j(x)}{x} + \sum_{k \geq 0} \frac{(-1)^k \zeta^{(j)}(-k)}{k!} x^k, \quad g_1 = h, \quad g_2 = h^2 + \zeta(2).$$

The subtraction P removes the singular product, both cross terms with $k = 0, 1$, and the constant analytic product. The remaining numerator is $O(x(1 + |\log x|^2))$ at zero. At infinity $f_1 f_2 = O(e^{-4x})$. This proves ordinary absolute convergence; normal convergence in a parameter polydisc justifies the differentiation used above.

Finally,

$$\Gamma(1-t) = 1 + gt + \frac{g^2 + h_2}{2}t^2 + \frac{g^3 + 3gh_2 + 2h_3}{6}t^3 + O(t^4).$$

Applying $\partial_A^2 \partial_B$ to the six elementary terms after $I(A, B)$ in (7.15), using this Taylor expansion, gives exactly E . This proves (7.14). $\square$

The representation (7.14) fixes the coordinate using only ordinary polylogarithms, their order derivatives, Gamma constants, and ordinary zeta derivatives. It is not an asserted reduction of Ξ to a finite algebra of standard constants. The splitting point 1 in (7.12) is part of the formula; changing it requires changing the explicitly integrated subtractions.

7.4 The formal relation space at arbitrary degree

Here a *formal degree- d freedom* means the difference of two homogeneous degree- d jets satisfying symmetry in A, B , the same degree of (7.3), and the same degree of (7.4). No other functional relation is imposed.

Theorem 7.4 (Exact homogeneous kernel). *For $d \geq 2$, every formal degree- d freedom has a unique representation*

$$\Delta R_d(A, B, C) = A(A - C)S(A, C) + B(B - C)S(B, C) + ABQ(A, B, C), \quad (7.16)$$

where S is a homogeneous symmetric polynomial of degree $d - 2$ in two variables and Q is a homogeneous polynomial of degree $d - 2$, symmetric in A, B . Therefore the dimension of this formal freedom is

$$\dim \mathcal{K}_d = \left\lfloor \frac{d}{2} \right\rfloor + \left\lfloor \frac{d^2}{4} \right\rfloor. \quad (7.17)$$

At degrees 0 and 1 the freedom is zero.

Proof. Put $D(A, C) = \Delta R_d(A, 0, C)$. The axis condition gives $D(0, C) = 0$, so $D = AV(A, C)$. The homogeneous Euler condition becomes

$$AC(V(A, C) + V(C, A)) = 0.$$

Hence V is antisymmetric in A, C , and $V(A, C) = (A - C)S(A, C)$, where S is symmetric. This proves the asserted boundary value.

Subtract the first two terms of (7.16) from ΔR_d . The difference is still symmetric in A, B and vanishes when $A = 0$ or $B = 0$. It is therefore divisible by AB ; its quotient is the required symmetric polynomial Q . This construction determines S from the boundary and then determines Q , proving uniqueness. Conversely, a direct substitution shows that every expression (7.16) satisfies all three homogeneous conditions.

The space of symmetric polynomials of degree $d - 2$ in two variables has dimension $\lfloor d/2 \rfloor$. The space of degree- $d - 2$ polynomials in three variables symmetric in the first two has dimension

$$\sum_{k=0}^{d-2} \left(\left\lfloor \frac{d-2-k}{2} \right\rfloor + 1 \right) = \left\lfloor \frac{d^2}{4} \right\rfloor.$$

For $d = 0, 1$, the divisibility argument forces S and Q to vanish. $\square$

The Tornheim derivative of order n uses degree $d = n - 1$ of R . Thus the first formal dimensions, for derivative orders $n = 1, \dots, 8$, are

$$0, \quad 0, \quad 2, \quad 3, \quad 6, \quad 8, \quad 12, \quad 15.$$

In particular the quartic three-coordinate count is exact under these stated relations. This count does not say that three arithmetically independent periods are necessary.

Corollary 7.5 (All-order cyclic cancellation and its exact image). *For $d \geq 2$, the image of the homogeneous freedom under*

$$\Delta R_d \mapsto c\Delta R_d(a, b, c) + a\Delta R_d(b, c, a) + b\Delta R_d(c, a, b)$$

is exactly abc times the space of symmetric homogeneous polynomials of degree $d - 2$ in a, b, c . Consequently its dimension is

$$p_{\leq 3}(d - 2) = \left\lfloor \frac{(d + 1)^2 + 3}{12} \right\rfloor.$$

Equivalently, at derivative order $n \geq 3$, cyclic symmetrization leaves exactly $\lfloor (n^2 + 3)/12 \rfloor$ formal coordinates.

Proof. The terms involving S in (7.16) cancel pairwise because $S(a, c) = S(c, a)$. The remaining expression is

$$abc(Q(a, b, c) + Q(b, c, a) + Q(c, a, b)).$$

The parenthesis is fully symmetric. Every symmetric homogeneous polynomial P occurs by choosing $Q = P/3$, so the map is onto. A basis of the symmetric space is $e_1^i e_2^j e_3^k$ with $i + 2j + 3k = d - 2$. Counting these triples gives the stated dimension. $\square$

Why the diagonal ceases to determine cyclic fifth derivatives. The preceding dimensions equal one at derivative orders three and four, which explains why the source's cubic cyclic formula and (7.8) each need just the corresponding diagonal derivative. At order five the dimension is two. This is not merely a count: the allowed deformation

$$\Delta R_4(A, B, C) = \frac{AB}{3}((A + B + C)^2 - 3(AB + BC + CA))$$

vanishes on the diagonal and preserves the axis and Euler relations, but changes the cyclic fifth derivative by

$$5! abc((a + b + c)^2 - 3(ab + bc + ca)).$$

Therefore the diagonal fifth derivative together with the named relations cannot determine every cyclic fifth ray. Additional analytic information is needed.

7.5 An exact symmetric generating germ

The preceding cyclic statement can be expressed without choosing coordinates for each degree.

Proposition 7.6 (Cyclic factorization). *There is a unique symmetric holomorphic germ $J(A, B, C)$ near the origin such that*

$$\begin{aligned} \sum_{\text{cyc}} T(A, B, C) &= 2(\zeta(A)\zeta(B) + \zeta(B)\zeta(C) + \zeta(C)\zeta(A)) \\ &\quad - \zeta(A + B) - \zeta(B + C) - \zeta(C + A) \\ &\quad + 2(\zeta(A) + \zeta(B) + \zeta(C)) + 1 + ABCJ(A, B, C). \end{aligned} \quad (7.18)$$

Proof. In the cyclic sum of (7.1), the two terms with denominator $A + C$ combine to $-\zeta(B - 1)$, and likewise for the other two denominators. Thus the cyclic sum is holomorphic and symmetric near the origin. On $B = 0$, the Euler relation gives

$$T(A, 0, C) + T(0, C, A) + T(C, A, 0) = 2\zeta(A)\zeta(C) - \zeta(A + C).$$

The explicit zeta expression in (7.18) has exactly the same restriction. By symmetry, their difference vanishes on all three coordinate planes. The convergent Taylor series is therefore divisible by ABC . Its quotient is holomorphic and symmetric, and uniqueness is immediate. $\square$

Let J_k denote the homogeneous degree- k Taylor polynomial of J . For every $n \geq 3$ and every direction whose pairwise sums are nonzero, differentiating the restricted identity gives the all-order formula

$$\begin{aligned} \sum_{\text{cyc}} W_{a,b,c}^{(n)}(0) &= 2 \sum_{\{u,v\} \in \{\{a,b\}, \{b,c\}, \{c,a\}\}} \sum_{j=1}^{n-1} \binom{n}{j} u^j v^{n-j} z_j z_{n-j} \\ &\quad - ((a + b)^n + (b + c)^n + (c + a)^n) z_n + n! abc J_{n-3}(a, b, c). \end{aligned} \quad (7.19)$$

The inner sum is symmetric in u, v , so the unordered-pair notation is unambiguous. All derivatives at -1 and all boundary Gamma coordinates have disappeared. At the first two orders of J ,

$$J_0 = \frac{\Omega}{2} + 3Lz_2 + 4z_3, \quad J_1(A, B, C) = \frac{A + B + C}{24}(\Lambda + 8Lz_3 - 12z_2^2 + 16z_4),$$

where $\Omega = W'''_{1,1,1}(0)$ is the incoming cubic diagonal coordinate. Thus (7.19) recovers the source's cubic identity and proves (7.8).

7.6 Verification, scope, and the next exact questions

The accompanying `check_quartic_tornheim.py` verifies (7.7), (7.8), the Euler specialization, and the elementary correction E by exact symbolic arithmetic. It independently constructs the homogeneous constraint matrices for degrees 0 through 10, computes their exact ranks and cyclic images, and checks them against the proved dimension formulas. An optional numerical mode evaluates the new coordinates through integrated Jonquière series and a Stirling-subtracted Gamma sum, and compares asymmetric fourth derivatives with an independent Mellin evaluation of T . Numerical discrepancies are diagnostics, not interval certificates.

The results answer the quartic directional-layer question and the formal homogeneous-dimension part of the all-order question in [3]. They do not evaluate Λ , Ξ , or the earlier cubic coordinates in a finite standard-constant algebra, and they do not settle S_6 or the revised S_8 . The Mellin and Jonquière ingredients are classical [6]; the quartic and formal kernel results are developments relative to the inspected repository snapshot. A global priority claim is not made.

The next concrete problems are to identify a useful independent analytic relation for the symmetric degree-two germ J_2 ; to construct an economical Gamma or Barnes representation for Ξ ; and to compare additional Tornheim functional relations with the kernel (7.16). Any improved coordinate count should specify exactly which extra relations are used.

8 Source audit, corrections of interpretation, and integration

8.1 What the audit establishes

The targeted review found no false theorem in the selected incoming Tornheim statements or in the harmonic continuation used here. It did find an attribution overlap that matters for presenting this contribution: the canonical manuscript already gives the polynomial Hurwitz transform and the complete-homogeneous coefficient mechanism for all Stieltjes moments. The precise locations are `chapters/07-integration.tex`, `stieltjes:sec:moments`, `stieltjes:thm:transform`, and `stieltjes:thm:moment` [1]. Accordingly, Corollary 6.4 is classified as a fully indexed arbitrary-endpoint restatement and extension, useful for the new harmonic reduction, rather than a new general principle.

The formal difference-equation input, the Espinosa–Moll generalized polygamma, and the classical Barnes identity are also credited explicitly. Likewise, the boundary Gamma representation for U in (7.10) is an existing theorem specialized to the derivative order needed here. These distinctions prevent an exact consequence from being counted twice as a separate research discovery.

The reviewed source files were taken from the fixed commit stated in Section 1. As an additional provenance check, the 44 inspected TeX files from the three newest incoming reports were compared byte-for-byte with the corresponding ZIP contents in that checkout; every comparison agreed. The packaged manifest records the nine archive names, their Git blob identifiers, and SHA-256 hashes. An integration based on a later repository revision should reconcile later changes explicitly.

8.2 Current conjectures must retain their current labels

The canonical conjectures `cycloquot:conj:S6` and `s8new:conj:S8` remain conjectural. The previously rejected S_8 coefficient vector is a different candidate from the revised one. A rejection of the former is not a rejection of the latter, and a numerical residual for either candidate is not an analytic proof.

Two different computational statements in the incoming collection also need to remain separate. The enlarged modular search reported in *Independent Orders* was interrupted before exhausting its matrix; that record alone establishes neither membership nor nonmembership. The later *Exact Resonant Identities* report establishes an exact obstruction for the complete Cayley ideal together with its specifically enumerated additional row family. That is a valid statement about the stated relation space. It does not exclude additional distribution, double-shuffle, or functional relations, and it is not a nonvanishing theorem for the actual period [3, 2].

The analogous qualification is essential for the present Theorem 7.4: the formal dimension is proved under symmetry, the C -axis identity, and the Euler-plane identity. Its exactness within that system does not imply arithmetic independence, nor does it claim to have imposed every possible Tornheim functional equation.

The earlier source corrections concerning Tin's equation (22), local polylogarithm estimates, and the rejected old S_8 vector were already reported before this contribution. They are not advertised here as newly discovered errors.

8.3 Specific normalization rules supported by the new proofs

The following are concrete instructions for using the formulas. They also identify the corrections required if one extends the existing calculus by a naive parameter substitution.

Retain the entire resonant amplitude. For the resolvent family the local term is $r_p(z)A_p(s)x^{s-1}$, with $A_p(s) = \prod_{j=1}^p (s-j)$. Its continued integral is $r_p A_p(s)/s$. Subtracting only its residue at zero leaves incorrect regular coefficients. The correct expression is (4.13), with the entire polynomial amplitude retained. At $p = 1$, failure to do so misses the explicit constant 1 in (4.17). This is a proved correction to that tempting extension, not an allegation that the pinned source already contains the incorrect resolvent formula.

Keep the endpoint coordinate. All endpoint finite parts in the article use x , with the cutoff specified in Section 4. A nonlinear endpoint substitution can change the finite constant of a logarithmic divergence. The resolvent's spectral answer is converted back to this stated coordinate before coefficient extraction. Cauchy boundary values in (5.9) are asserted for the ordinary $p = 0$ integral; no unproved transport of divergent higher-derivative finite parts is implicit.

Do not differentiate a fixed index as though it were free. The deletion theorem is proved on the hyperplane where a specified index equals $-m$. It permits every derivative in the remaining independent free indices, because it is an entire identity in those indices. It does not determine the normal derivative transverse to the eliminated-index hyperplane. That additional information is a separate research problem.

Keep both Bernoulli endpoint conventions. The strict-gap deletion rule uses $B_1(0) = -1/2$ and $B_1(1) = 1/2$. Identifying them incorrectly deletes the last term in

$$\mathcal{H}(s, 0, t) = \mathcal{H}(s-1, t) - \mathcal{H}(s, t-1) - \mathcal{H}(s, t).$$

The verification suite deliberately corrupts that sign and confirms that an independent finite nested sum detects it.

Restrict individual Tornheim rays before specialization. The individual fourth derivative in (7.7) requires $(a + c)(b + c) \neq 0$. A sum of cyclic germs can extend across planes where individual terms are singular. The holomorphic cyclic identity (7.18) has that extension; its representation as a sum of individually restricted derivatives is used only when every pairwise direction sum is nonzero.

Specify what is centered and what is simultaneous. The involution in Theorem 2.1 changes all even harmonic tails simultaneously. It does not require each tail to occur to an even power separately. For example $(\zeta(2) - H_n^{(2)})(\zeta(4) - H_n^{(4)})$ qualifies. Conversely, a raw even-order harmonic number does not qualify: $H_n^{(2)}$ produces the pole $-1/s$ at $s = 0$. This prevents two opposite misreadings of the classification.

8.4 Proposed integration into the research collection

The material separates naturally into four independently readable additions: the centered generalized-harmonic classification and sums; rational cotangent transforms and balanced primitives; arbitrary-depth harmonic deletion and the Lerch generator; and quartic and higher Tornheim jets. Each has distinct label prefixes in the supplied TeX.

The incoming source questions can be annotated with the exact answers in Table 1. Q11 receives a complete classification. Q8 receives the stated rational sector. Q9 receives the balanced primitive ladder for these kernels, while a closed formula for every unbalanced Barnes product is still open. The quartic directional question receives a full normal form, and the all-order question receives an exact answer for its specified formal relation system.

No existing conjecture should be promoted to a theorem unless it is exactly one of these proved statements. The package includes a claim ledger, proposed placement notes, source hashes, and separate exact and numerical verification records so that this distinction can survive later merging and renumbering.

9 Further research: precise next questions

The proved results suggest several exact research problems. They are listed with the additional information needed for a solution. None is asserted on numerical evidence alone.

9.1 Centered harmonic values and global spectral derivatives

R1. Evaluate mixed even-tail moments in a controlled algebra. The invariant-ring theorem includes every product

$$(\zeta(2i) - H_n^{(2i)})(\zeta(2j) - H_n^{(2j)})$$

and its products with odd-order harmonic numbers and powers of H_n . Determine finite formulas for their centered values at $s = -2r$, beginning with $i = 1, j = 2$. The proof of Theorem 3.1 suggests finite summation by parts followed by Bernoulli coefficient extraction. The target should specify an ordinary or multiple-zeta coordinate space at each weight, with an explicit account of the terms that survive. A useful strong result would sum an entire family in one convergent generator, as in (3.15).

R2. Classify shifts other than one half. For a fixed positive rational $a \neq 1/2$, determine all polynomials P whose principal parts vanish on the even nonpositive spectral lattice. Proposition 2.2 gives every finite obstruction, but the centered involution no longer has its simple form. The key question is whether the infinitely many Bernoulli constraints admit a finite algebraic description. A classification of simultaneous conditions at a and $1 - a$ may be more natural than a classification at one noncentral shift.

R3. Separate local regularity from global first derivatives. The absence of a principal part is a local asymptotic statement. The value of the first regular spectral derivative also contains a global remainder integral. Find polynomial combinations $P_1, \dots, P_k$ and exact coefficients for which these remainder kernels cancel at a nonpositive even center. A satisfactory answer is an identity of kernels or a normally convergent sum, not a numerical relation among already evaluated derivatives. The centered parity classification narrows the possible local terms and should reduce this search.

9.2 Beyond rational cotangent kernels

R4. Complex powers with moving endpoint exponents. Extend the completed generator to

$$\text{FP}_x \int_0^1 \gamma_m^{(p)}(x)(K(x) - z)^{-\lambda} dx, \quad z \notin \mathbb{R}.$$

The local exponent is $\lambda - p - 1$ before logarithmic differentiation. Thus integer crossings of λ create new resonances not present in a fixed repeated resolvent. First choose a logarithm of $K(x) - z$ continuously along the interval, then prove a joint meromorphic generator and identify each full resonant amplitude before expanding either parameter. The rational integer cases of Corollary 4.2 are exact normalization data for this problem.

R5. Products of continuous polygamma orders. The transform (5.2) is linear in the generalized function $\Psi(v, x)$. For products such as $\Psi(v, x)\Psi(w, x)$, Fourier convolution introduces ordered or Tornheim-type coordinates. Determine which symmetric or reflected combinations reduce to ordinary polylogarithmic order derivatives and which genuinely retain a multivariate zeta coordinate. The useful deliverable is an explicit joint generator together with the contact terms on every resonant hyperplane.

R6. Unbalanced Barnes functions with fixed normalization. The balanced ladder gives combinations of $\zeta'(-r, x)$ and $\zeta(-r, x)$, and hence normalized Barnes combinations at the first levels. Seek individual resolvent transforms of $\log G(x)$ and higher multiple Gamma functions, with every polynomial-in- x companion term and normalization constant stated. A functional equation alone does not determine the function without the customary asymptotic normalization. One concrete starting point is to combine (5.7) with transforms containing $x \log \Gamma(x)$, whose Fourier coefficients can be computed by one integration by parts.

9.3 Entire harmonic interpolation at more than one free order

R7. A jointly convergent generator for several free orders. The canonical gap polynomial is already explicit for k surviving free indices. Sum arbitrary zero blocks before, after, and between these indices into a useful multivariate analytic kernel. The coefficient identity alone is insufficient: prove a domain in all decoration variables, the cancellation of simultaneous Lerch branches, and normal convergence of independent spectral derivatives. The one-free-order formula (6.20) supplies a stringent marginal specialization for any proposed answer.

R8. The first transverse derivative at a deleted index. Differentiate a word in an index that will subsequently be specialized to $-m$. Its lattice summand acquires $-x^m \log x$, so a Bernoulli polynomial no longer represents the gap sum. Identify a finite family of new gap functions, perhaps in normalized Hurwitz-zeta derivatives, closed under the same concatenation and inversion argument. The first target is a fully proved formula for

$$\partial_u \mathcal{H}_{a,b}(s, u, t)|_{u=0}$$

at arbitrary independent s, t . This would extend the present theorem in a genuinely transverse direction.

R9. Rational endpoints and exact character projections. Combine the explicit endpoint coefficient formulas with finite character decompositions at rational a, b . The expected output is an algorithm expressing the specified harmonic jets and polynomial Stieltjes moments in Dirichlet L -derivatives and conductor logarithms. The finite reduction should identify the character parity, the normalization of the endpoint difference, and every logarithmic derivative of the conductor factor. This is particularly suited to exact symbolic verification.

9.4 The next Tornheim coordinates and the Gaussian conjectures

R10. Determine the second-degree symmetric germ. The polynomial J_2 in (7.18) has two independent formal coordinates under the named relations. The diagonal fifth derivative provides only one linear combination. Obtain one additional independent analytic identity or a normalized ordinary convergent integral that determines the other combination. The explicit deformation following Corollary 7.5 proves why manipulating the existing diagonal and Euler-plane relations alone cannot finish this task.

R11. Compare additional functional relations with the exact kernel. The representation (7.16) is an explicit target against which another Tornheim relation can be tested. Expand a proposed functional relation at the origin, compute its action on the S and Q components, and state exactly which directions it removes. This would turn a collection of individual ray identities into a systematic all-order reduction theory. Any improved dimension statement must list the added relations; numerical dependence among sampled coordinates is not a substitute.

R12. Give an independent Gamma or Barnes representation for Ξ . The ordinary integral (7.14) is explicit, but it does not yet explain whether the mixed coordinate belongs to a smaller familiar constant algebra. A useful next theorem would derive a different Gamma or Barnes representation and prove its equality with the polylogarithmic integral. Such an independent representation may reveal functional symmetries or exact cancellations that are hidden in either form. Simply changing the splitting point in the same subtracted integral does not provide new information.

R13. Find a functional bridge to the actual S_6 and revised S_8 candidates. The short Gaussian reductions remain important open problems in the source. The completed bounded relation obstruction shows that a proof must use information beyond the particular imposed Cayley and extra-row system. A constructive next step is an explicit one-parameter identity whose Taylor coefficient is the target mixed harmonic sum and whose second evaluation uses the proposed Gaussian polylogarithmic constants. All boundary terms, algebraic arguments, and logarithm branches must be fixed before specialization. The rational and Lerch generators in this article offer possible tools, but no such bridge to either named candidate has been proved here.

9.5 A practical order of attack

The most immediate exact projects are R1, R7, and R8: each starts with a finite normal form or a convergent kernel already supplied here. R10 and R11 are the natural next steps for the directional-zeta program, because the obstruction identifies precisely what new information is missing. R4 and R5 require a more substantial joint resonance analysis.

For formal verification, the first useful units are the Bernoulli deletion identities, the canonical gap normal form, the finite principal-part coefficients, the quartic coefficient map, and the kernel decomposition (7.16). These are exact finite algebraic statements. Their analytic interfaces should then be formalized separately: normal convergence, endpoint subtraction, and the specified continuation theorems. This separation makes a future machine-checked development concrete without pretending that the present numerical diagnostics already supply one.

10 Verification and reproducibility

10.1 Analytic proofs, finite algebra, and numerical evidence

The proofs in the article establish the stated infinite families. Finite symbolic tests check the coefficient algebra and the implementation of the formulas. Numerical tests compare independent representations and help detect phase, endpoint, or transcription errors. A small floating-point discrepancy is not an interval enclosure and is not used as a proof of an identity.

The complete numerical records are supplied as JSON rather than rounded tables alone. They record working precision, truncations, directions or endpoints, and the actual observed discrepancies. For a reproducible source baseline, the package also includes the archive and canonical-file hashes and a claim ledger distinguishing new developments, inherited inputs, explicit restatements, and unresolved questions.

10.2 Centered harmonic and trigamma-square checks

The file `code/verify_harmonic_parity.py` verifies representative invariant and anti-invariant polynomials through inverse degree 16, the general-shift expansions at $a = 1/3$ and $a = 7/5$, and the formal trigamma shift equation through the stated truncation. It independently checks the two finite-summation constant terms for $j = 1, \dots, 8$. These finite tests supplement the all-order injectivity proof; they do not replace it.

At 110-digit working precision, the values $\mathcal{T}(-2r)$ for $r = 0, \dots, 6$ were compared with a direct finite trigamma sum plus an Euler–Maclaurin tail. The two cutoff choices were $(N, J) = (64, 38)$ and $(96, 45)$. The largest discrepancy of the tighter-cutoff value from the theorem was approximately $1.51 \cdot 10^{-85}$; the largest difference between the two cutoff calculations was $9.28 \cdot 10^{-69}$. At $t = 0.2, 0.7, 1.1$, the polylogarithmic generator was also compared with the Bernoulli moment series and the ordinary integral generator. The largest moment-series discrepancy was below $4.46 \cdot 10^{-95}$. No rigorous remainder enclosure is inferred from these comparisons.

10.3 Resolvent and continuous-order checks

The file `code/verify_resolvent.py` performs 55 exact assertions on rational local coefficients, the Gamma contact polynomial, its harmonic derivative, and the threshold where that correction vanishes. The completed numerical run has 28 checks at 55-digit working precision. They include upper- and lower-half-plane parameters, twelve pointwise polygamma finite parts, the explicit trigamma correction, the first three balanced primitives at two parameters, two convergent γ'_1 squared-resolvent integrals, four direct Stieltjes integrals, and two continuous polygamma orders.

A special-value regression is also included.

For the ordinary and local-subtraction checks assigned tolerance 10^{-42} , the largest observed discrepancy was below $3.31 \cdot 10^{-53}$. The endpoint-sensitive complex order $v = 0.3 + 0.2i$ had discrepancy approximately $8.04 \cdot 10^{-41}$, within its separately stated 10^{-34} tolerance. An additional independently implemented integration-by-parts check in `code/verify_resolvent_contact.py` tests nine polygamma contact values at 100 digits, with largest discrepancy below $5.27 \cdot 10^{-97}$.

The initial numerical implementation used finite differences of the Hurwitz function at spectral order zero. At the quadrature node $x = 7/8$, that evaluation lost precision even though the mathematical formula was exact. The final program uses the log-Gamma identity at this order and native Hurwitz order derivatives with an exact endpoint shift elsewhere. The regression at $7/8$ is retained, and no tolerance was relaxed. This is an implementation correction, not a change to the theorem.

10.4 Harmonic elimination, Lerch continuation, and primitives

The exact reducer `code/reduce_harmonic.py` is exercised by `code/verify_harmonic.py`. Its recorded checks comprise 26 words, 78 comparisons among different local deletion orders and the canonical gap form, 52 independent finite nested sums, 327 coefficient degree conditions, seven zero-insertion identities, and 36 complete-harmonic coefficient identities. A deliberately corrupted strict-gap sign is detected.

At 60-digit precision, six arbitrary noninteger-endpoint continuations and one free-order derivative were evaluated by an independent Euler–Maclaurin procedure. Their largest observed residual was below $1.43 \cdot 10^{-33}$. The zero-decoration generating function, truncated at total degrees 10, 14, 18, was compared with separate Lerch evaluations; the residuals were approximately $3.53 \cdot 10^{-20}$, $1.53 \cdot 10^{-26}$, and $7.82 \cdot 10^{-33}$, respectively.

An ordinary Mellin integral checks the Lerch formula in its honest convergence domain. Three normalized polynomial primitives and five Stieltjes quadratures give additional independent checks, with discrepancies at approximately 59–62 decimal places. A Python reciprocal initially formed at machine precision for an exact integer input was replaced by arbitrary-precision division; the two affected cases were rerun and the actual corrected records retained.

10.5 Tornheim coefficient and Mellin checks

The file `code/check_quartic_tornheim.py` checks the full quartic coefficient formula, its cyclic form, diagonal and Euler specializations, and the elementary correction in the mixed coordinate Ξ by exact symbolic arithmetic. It independently constructs the symmetry, axis, and Euler-plane constraint matrices for remainder degrees 0 through 10, computes their ranks and cyclic images, and compares them with the dimension formulas.

The numerical mode uses 70-digit working precision, 68 integrated Jonquière terms, an incomplete-Gamma cutoff of 165, and a 16-node fourth-derivative formula with step 0.000015. The independent Mellin comparisons were:

Direction (a, b, c)	Absolute discrepancy
$(1, 2, 3)$	$3.739835213 \cdot 10^{-37}$
$(2, 3, 1)$	$2.075573689 \cdot 10^{-38}$
$(1, -2, 3)$	$2.871059279 \cdot 10^{-39}$

Reducing the finite-difference step by a factor of ten from an initial run decreased the first discrepancy by the expected scale for the retained truncation error. The delivered record is the completed tighter-step run. The exact proof does not depend on that numerical tuning.

10.6 How to rebuild and inspect the package

The standalone file `Harmonic_Parity_and_Resolvent_Identities.tex` contains the complete article, including its bibliography. It requires only a standard LaTeX installation with the listed packages and can be compiled without network access. The modular `article.tex`, `preamble.tex`, and `sections/` files are also retained for integration.

Run `python3 build.py` to regenerate the standalone source from the modules and compile the PDF. Run `python3 code/verify_all.py` for the routine exact and moderate numerical checks, or add `-full` to rerun the slower independent Stieltjes, Lerch, and Tornheim comparisons. The individual scripts remain usable separately. Python dependencies and versions are recorded in the package.

The PDF was compiled and inspected for unresolved references, unavailable glyphs, overfull displays, and page layout. The independent mathematical review notes are supplied as review artifacts. They are an additional check by separate derivations, not external peer review or a machine-checked proof.

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